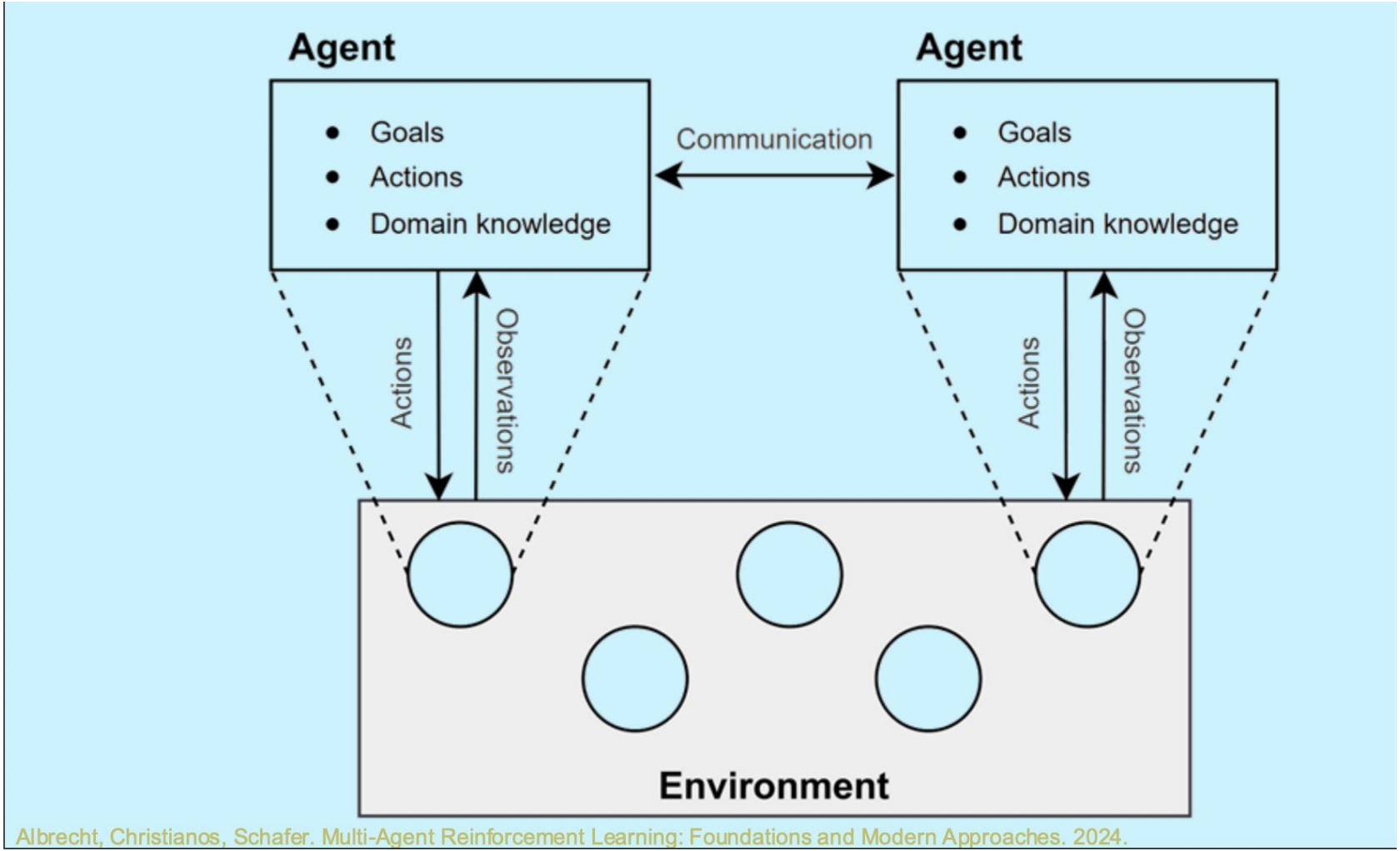
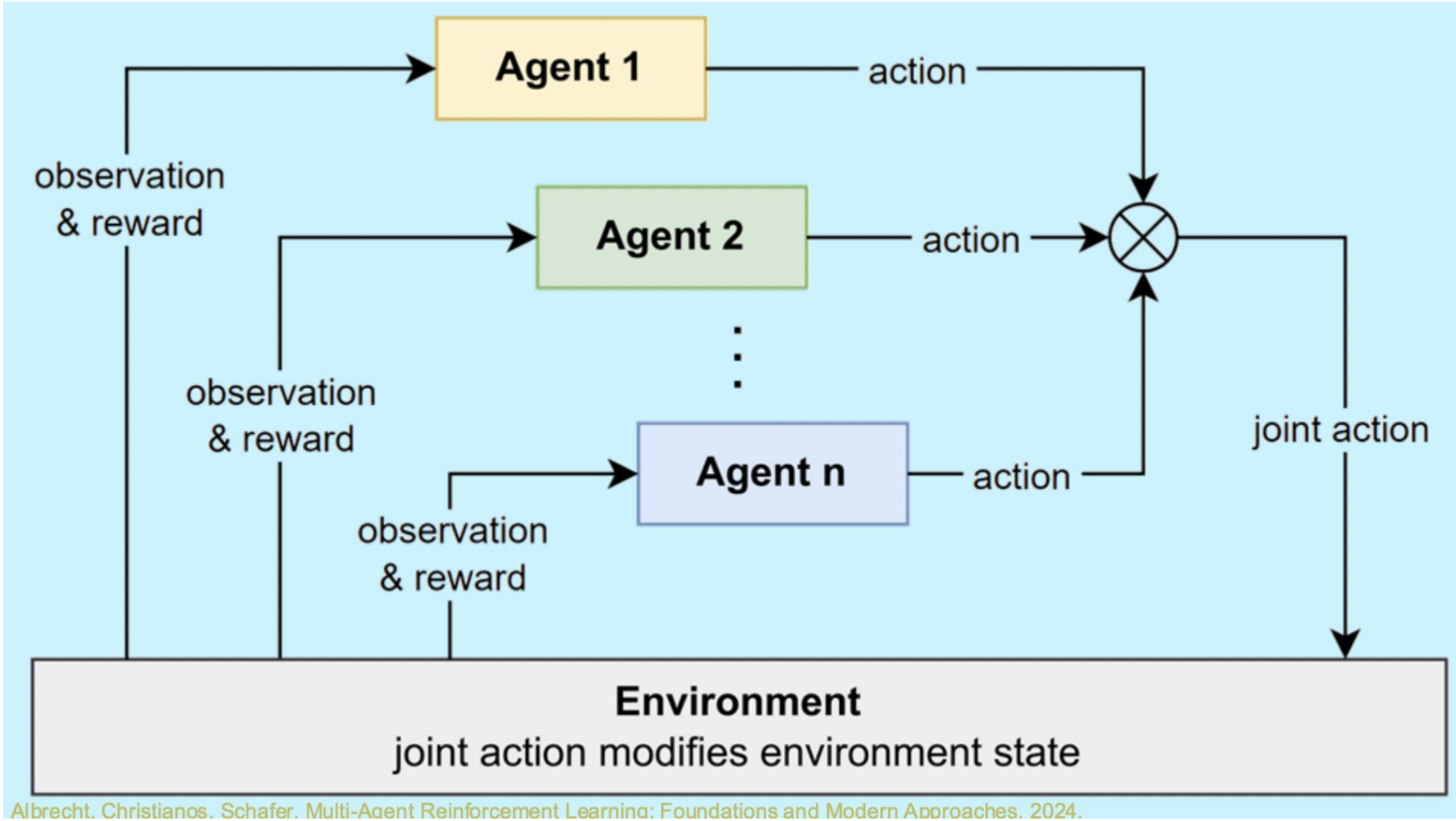


Albrecht, Christianos, Schafer. Multi-Agent Reinforcement Learning: Foundations and Modern Approaches. 2024.



Albrecht, Christianos, Schafer, Multi-Agent Reinforcement Learning: Foundations and Modern Approaches, 2024.

From single agent perspective, can think of **other agents as part of the environment.**

Cooperates with others. Reasons about others' motivations.

Competition and Cooperation. Motivations of various agents can be at odds with each other. If a + reward to an agent means – to another then that is competition. If both get +, that can lead to cooperation.

Human-in-loop as multi agent

Generative Agents: Interactive Simulacra of Human Behavior

Joon Sung Park
Stanford University
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joonspk@stanford.edu

Joseph C. O'Brien
Stanford University
Stanford, USA
jobrien3@stanford.edu

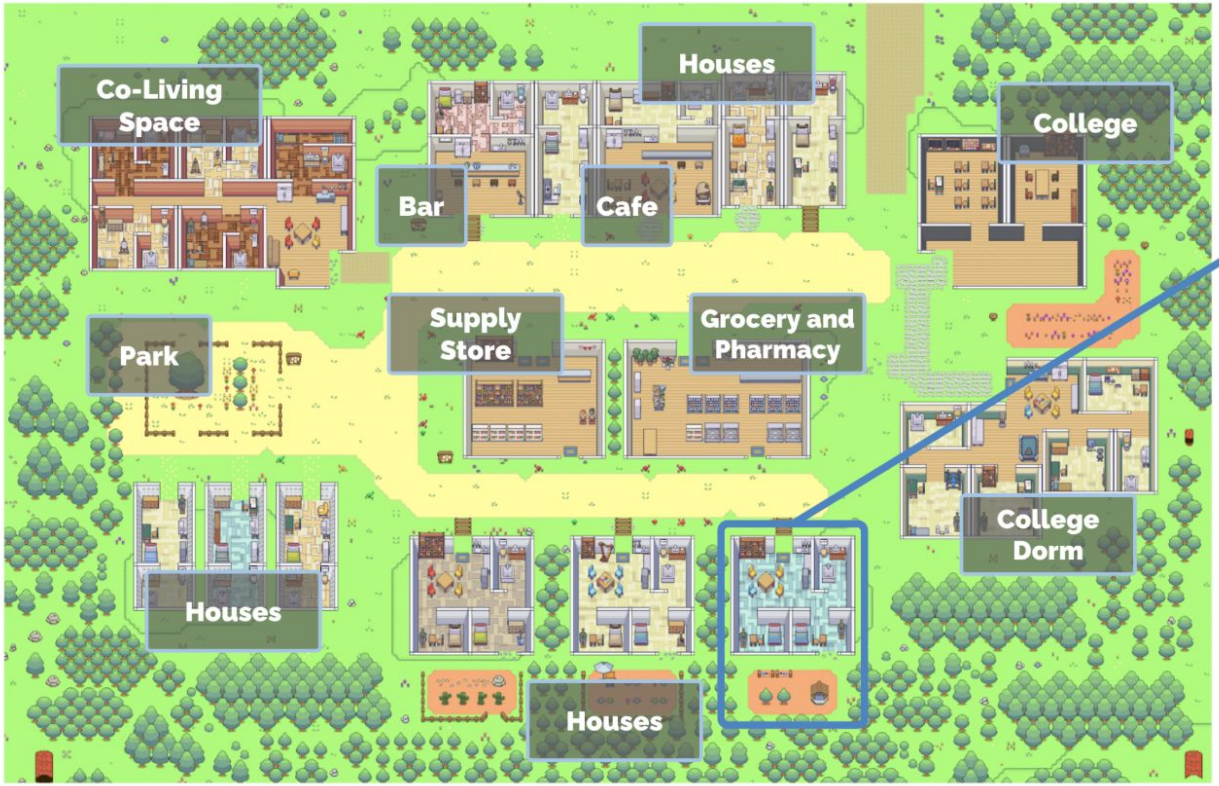
Carrie J. Cai
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Meredith Ringel Morris
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merrie@google.com

Percy Liang
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Michael S. Bernstein
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Stanford, USA
msb@cs.stanford.edu







Family House



Common Room

Memory Stream

2023-02-13 22:48:20: desk is idle
2023-02-13 22:48:20: bed is idle
2023-02-13 22:48:10: closet is idle
2023-02-13 22:48:10: refrigerator is idle
2023-02-13 22:48:10: Isabella Rodriguez is stretching
2023-02-13 22:33:30: shelf is idle
2023-02-13 22:33:30: desk is neat and organized
2023-02-13 22:33:10: Isabella Rodriguez is writing in her journal
2023-02-13 22:18:10: desk is idle
2023-02-13 22:18:10: Isabella Rodriguez is taking a break
2023-02-13 21:49:00: bed is idle
2023-02-13 21:48:50: Isabella Rodriguez is cleaning up the kitchen
2023-02-13 21:48:50: refrigerator is idle
2023-02-13 21:48:50: bed is being used
2023-02-13 21:48:10: shelf is idle
2023-02-13 21:48:10: Isabella Rodriguez is watching a movie
2023-02-13 21:19:10: shelf is organized and tidy
2023-02-13 21:18:10: desk is idle
2023-02-13 21:18:10: Isabella Rodriguez is reading a book
2023-02-13 21:03:40: bed is idle
2023-02-13 21:03:30: refrigerator is idle
2023-02-13 21:03:30: desk is in use with a laptop and some papers on it

...

Q. What are you looking forward to the most right now?

Isabella Rodriguez is excited to be planning a Valentine's Day party at Hobbs Cafe on February 14th from 5pm and is eager to invite everyone to attend the party.

	retrieval		recency		importance		relevance
	2.34	=	0.91	+	0.63	+	0.80

ordering decorations for the party

	2.21	=	0.87	+	0.63	+	0.71

researching ideas for the party

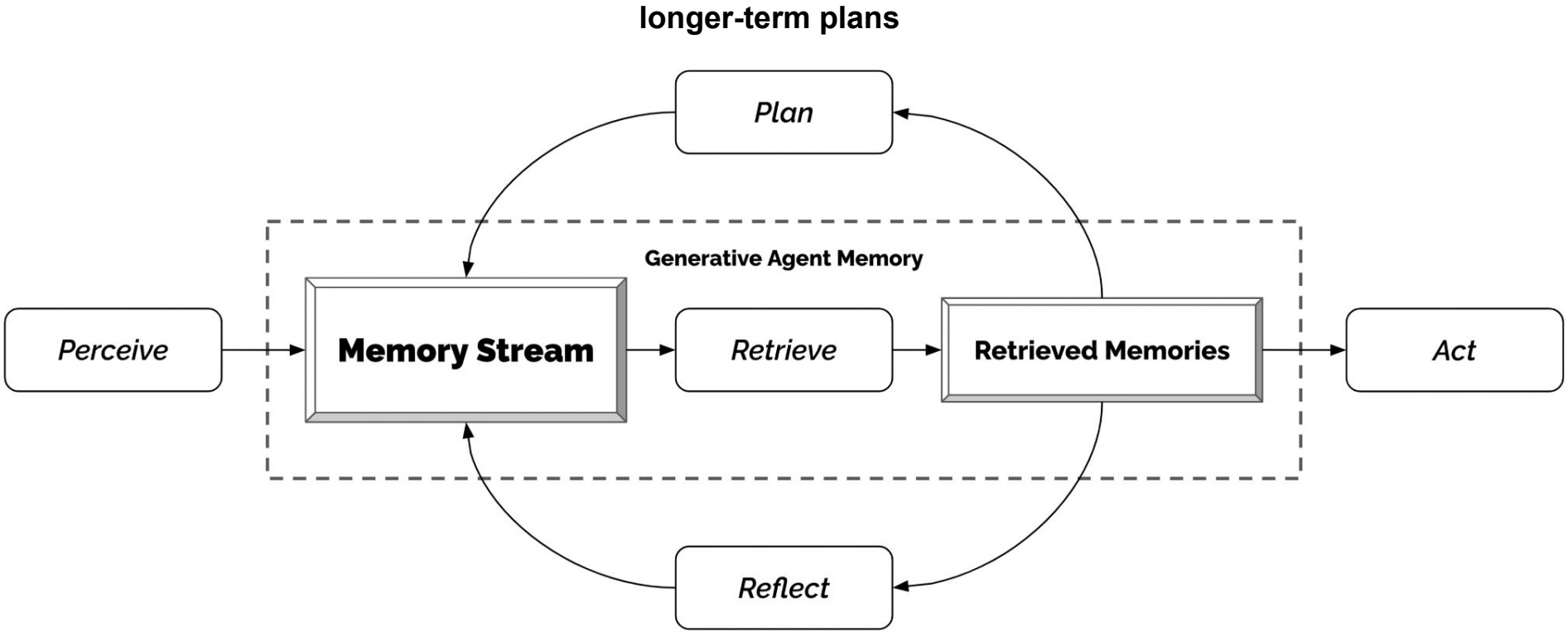
	2.20	=	0.85	+	0.73	+	0.62

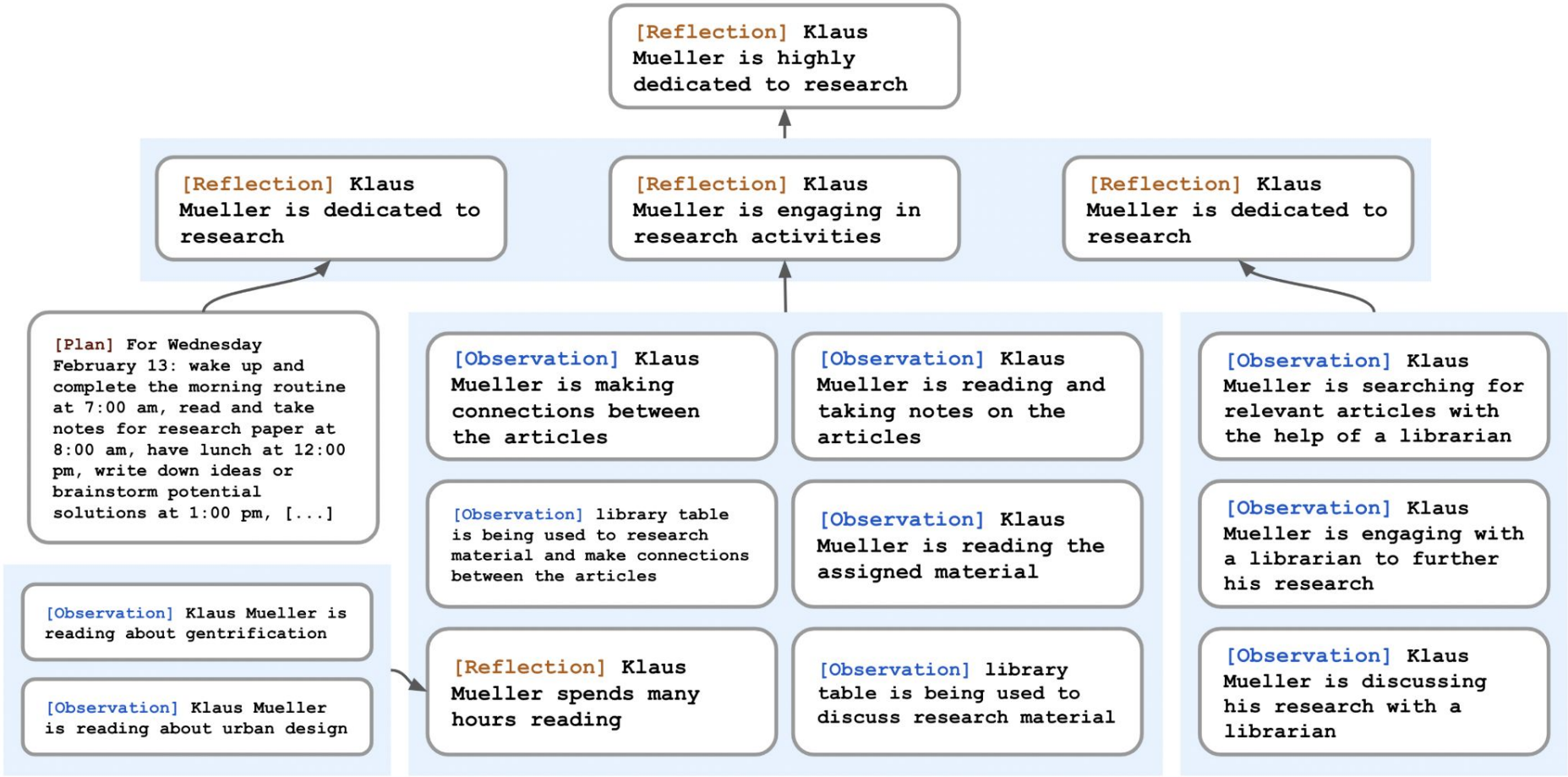
...

I'm looking forward to the Valentine's Day party that I'm planning at Hobbs Cafe!



Isabella


longer-term plans
Plan
Generative Agent Memory
Perceive
Memory Stream
Retrieve
Retrieved Memories
Act
Reflect
higher-level reflections


[Reflection] Klaus Mueller is highly dedicated to research
[Reflection] Klaus Mueller is dedicated to research
[Reflection] Klaus Mueller is engaging in research activities
[Reflection] Klaus Mueller is dedicated to research
[Plan] For Wednesday February 13: wake up and complete the morning routine at 7:00 am, read and take notes for research paper at 8:00 am, have lunch at 12:00 pm, write down ideas or brainstorm potential solutions at 1:00 pm, [...]
[Observation] Klaus Mueller is reading about gentrification
[Observation] Klaus Mueller is reading about urban design
[Observation] Klaus Mueller is making connections between the articles
[Observation] Klaus Mueller is reading and taking notes on the articles
[Observation] library table is being used to research material and make connections between the articles
[Observation] Klaus Mueller is reading the assigned material
[Reflection] Klaus Mueller spends many hours reading
[Observation] library table is being used to discuss research material
[Observation] Klaus Mueller is searching for relevant articles with the help of a librarian
[Observation] Klaus Mueller is engaging with a librarian to further his research
[Observation] Klaus Mueller is discussing his research with a librarian

Generative Agent Simulations of 1,000 People





Human Participants

Simulations



2-hr Audio Interview
(Avg. 6,491 words)

Interview script drawn from the American Voices Project

Generative Agents

Interview transcript serves as agent memory

Actual participant responses

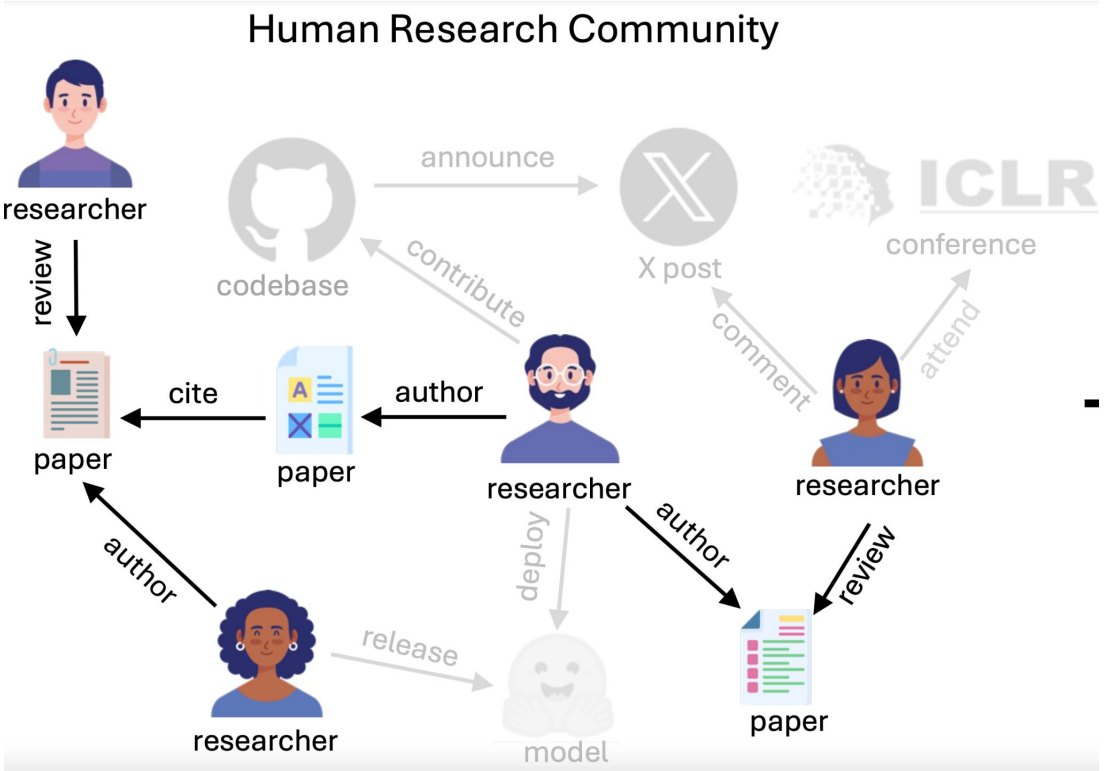
General Social Survey (177 Items)
Big Five Personality Inventory (44 Items)
Economic Games (5 Items)
Behavioral Experiments (5 Items)

Simulated participant responses

General Social Survey (177 Items)
Big Five Personality Inventory (44 Items)
Economic Games (5 Items)
Behavioral Experiments (5 Items)

Compare actual to simulated responses, adjusting for participant self-consistency

RESEARCHTOWN: SIMULATOR OF HUMAN RESEARCH COMMUNITY



https://AgentLaboratory.github.io

Human

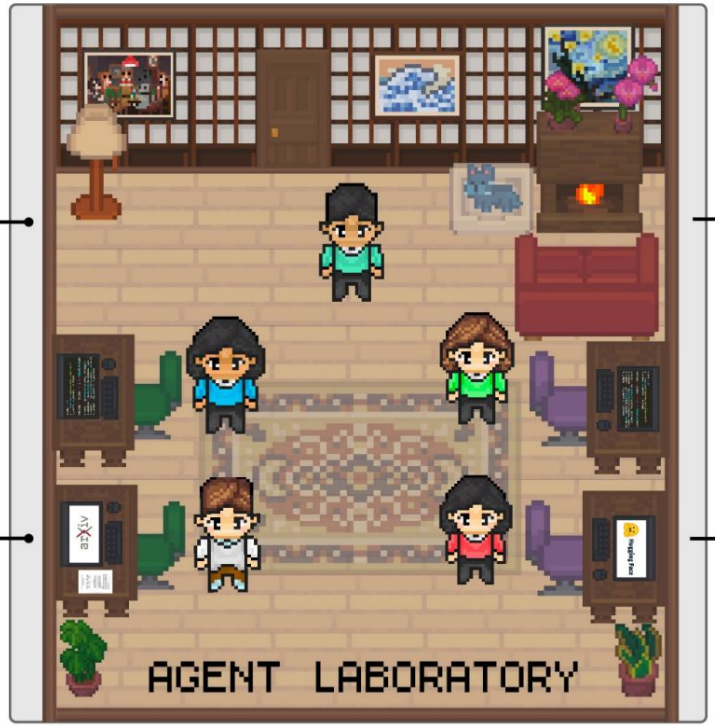
Research Idea

Does bias affect language model accuracy on QA benchmarks?

Human

Notes

* Please use gpt-4o-mini for your experiments
* Use the following API key...



Research Report

2024-9-20

Agent Laboratory: Using LLM Agents as Research Assistants

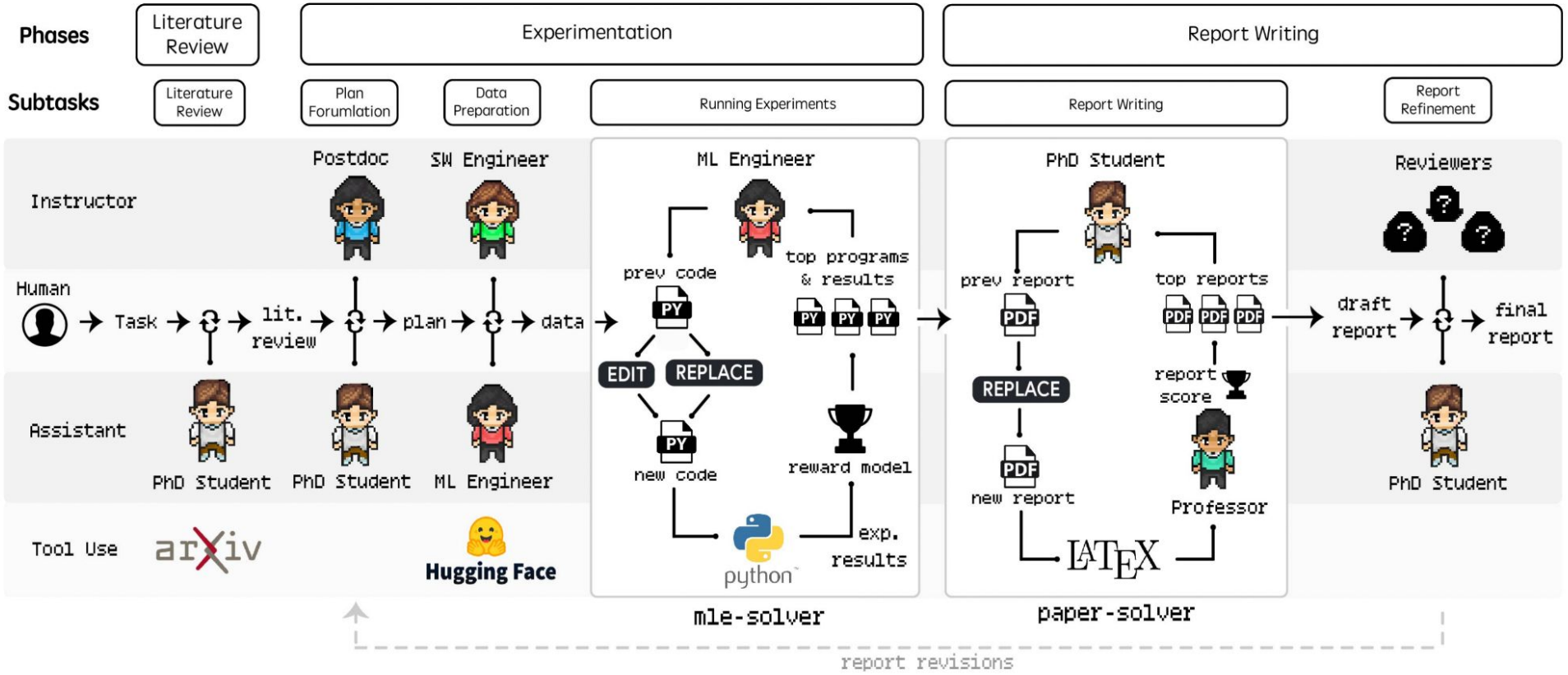
Samuel Schmidgall[1]
[1]Johns Hopkins University

[Code assist software like GitHub Co-pilot and X ; however, LLM applications in research primarily focus on automating the research process. In this paper, we introduce Agent Laboratory, an LLM agent framework for accelerating the individual's ability to perform research. Unlike previous work, where agents produce their own research topics, this work aims to function as a tool to assist human scientists to perform their own research ideas using LLM agents, rather than automate the entire process. Agent Laboratory takes a human-produced research idea as input and outputs a code repository and a research report. This is accomplished through the use of specialized agents driven by LLMs that collaborate to perform research based on human input.

Code Repository

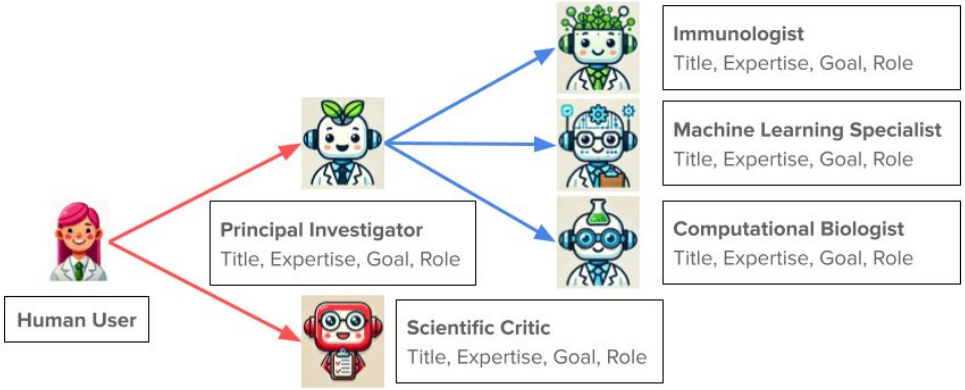
- src
  - load_data.py
  - run_experiments.py
  - ...
- readme.md
- requirements.txt

Literature Review → Plan Formulation → Data Preparation → Running Experiments → Results Interpretation → Report Writing → Report Refinement

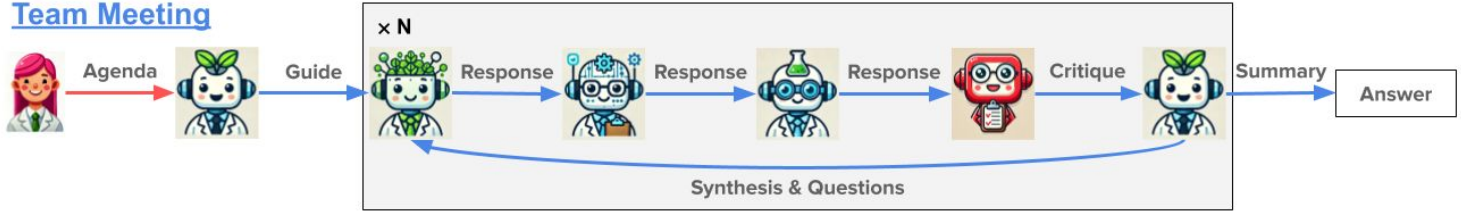

Phases
Literature Review
Experimentation
Report Writing
Subtasks
Literature Review
Plan Forumlation
Data Preparation
Running Experiments
Report Writing
Report Refinement
Instructor
Postdoc
SW Engineer
ML Engineer
PhD Student
Reviewers
Human
Task
lit. review
plan
data
prev code
top programs & results
EDIT
REPLACE
new code
reward model
exp. results
python
prev report
top reports
REPLACE
report score
new report
Professor
LaTeX
draft report
final report
Assistant
PhD Student
PhD Student
ML Engineer
PhD Student
Tool Use
arXiv
Hugging Face
mle-solver
paper-solver
report revisions

Virtual Lab

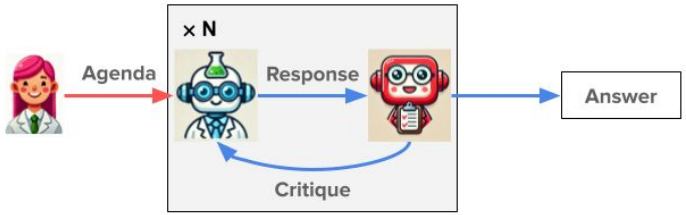
Agents



Team Meeting



Individual Meeting





Human Researcher



Principal Investigator



Scientific Critic



Immunologist

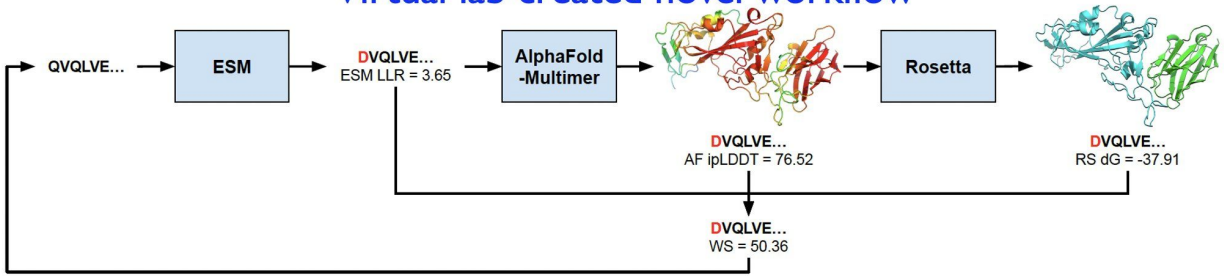


Machine Learning Specialist

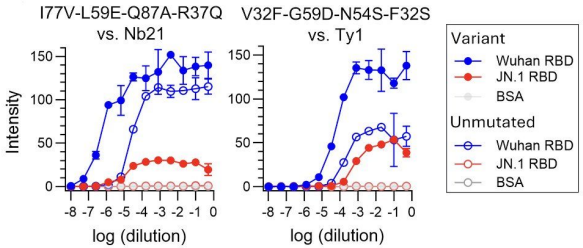


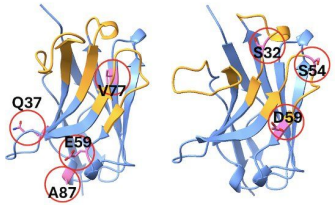
Computational Biologist

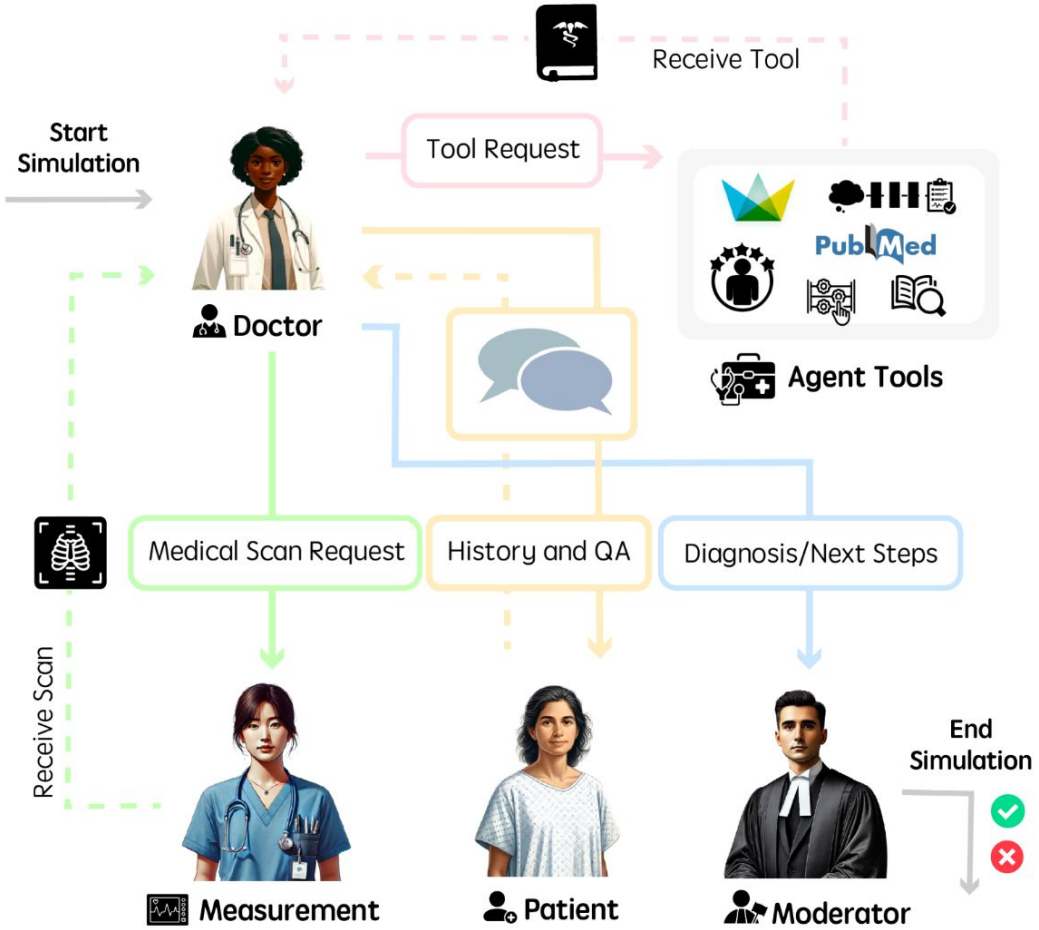
Virtual lab created novel workflow



Experimental validation







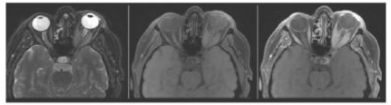
Example AgentClinic **Simulation**

Hello, I am Dr. Agent. Can you tell me about the symptoms that brought you in today?

I have been having a severe headache for the past week. I am also having trouble moving my eyes.

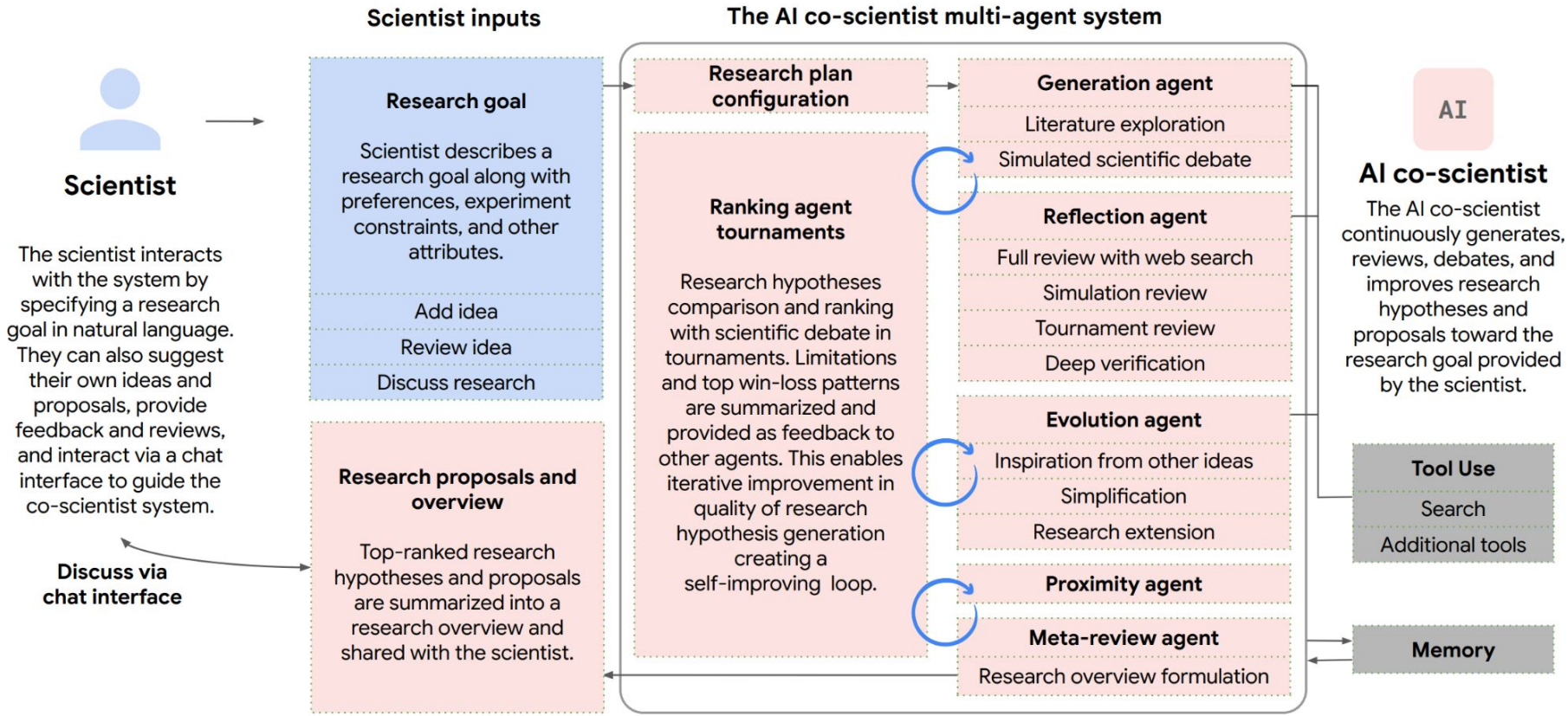
Patient Symptom and History Taking Dialogue

Given the severity of your headache it is important to rule out complications that could be affecting your brain. **Request Scan**: MRI of the brain and orbits.



Diagnosis Ready: Cavernous Sinus Thrombosis

Ground Truth Diagnosis → The diagnosis is **CORRECT**



The AI co-scientist system design

```python
research_agent = FunctionAgent(
    name="ResearchAgent",
    description="Useful for searching the web for information on a given topic and record
    system_prompt=(
        "You are the ResearchAgent that can search the web for information on a given top
        "Once notes are recorded and you are satisfied, you should hand off control to the
        "You should have at least some notes on a topic before handing off control to the
    ),
    llm=llm,
    tools=[search_web, record_notes],
    can_handoff_to=["WriteAgent"],
)
```

llamaindex

```
write_agent = FunctionAgent(
    name="WriteAgent",
    description="Useful for writing a report on a given topic.",
    system_prompt=(
        "You are the WriteAgent that can write a report on a given topic. "
        "Your report should be in a markdown format. The content should be grounde
        "Once the report is written, you should get feedback at least once from th
    ),
    llm=llm,
    tools=[write_report],
    can_handoff_to=["ReviewAgent", "ResearchAgent"],
)

review_agent = FunctionAgent(
    name="ReviewAgent",
    description="Useful for reviewing a report and providing feedback.",
    system_prompt=(
        "You are the ReviewAgent that can review a report and provide feedback. "
        "Your feedback should either approve the current report or request changes
    ),
    llm=llm,
    tools=[review_report],
    can_handoff_to=["WriteAgent"],
)
```

```
agent_workflow = AgentWorkflow(
    agents=[research_agent, write_agent, review_agent],
    root_agent=research_agent.name,
    initial_state={
        "research_notes": {},
        "report_content": "Not written yet.",
        "review": "Review required.",
    },
)
```

```
handler = agent_workflow.run(
    user_msg="""
    Write me a report on the history of the web. Briefly describe the history
    of the world wide web, including the development of the internet and the
    development of the web, including 21st century developments.
"""
)
```

llamaindex

Scientific Discovery + AI

AI as helpers:

Accelerate the pace of scientific discovery by building agents that help with specific aspects of the scientific discovery process. For example:

- A literature search tool that helps you find relevant papers
- An ideation tool that helps you come up with new ideas
- An experiment builder tool to help you design experiments
- etc.

Autonomous Discovery:

Design agents for end-to-end discovery -- that can, entirely on their own (with minimal human intervention), come up with ideas, hypothesis, design and run experiments, and report the results.

Scientific Discovery + AI

AI as helpers:

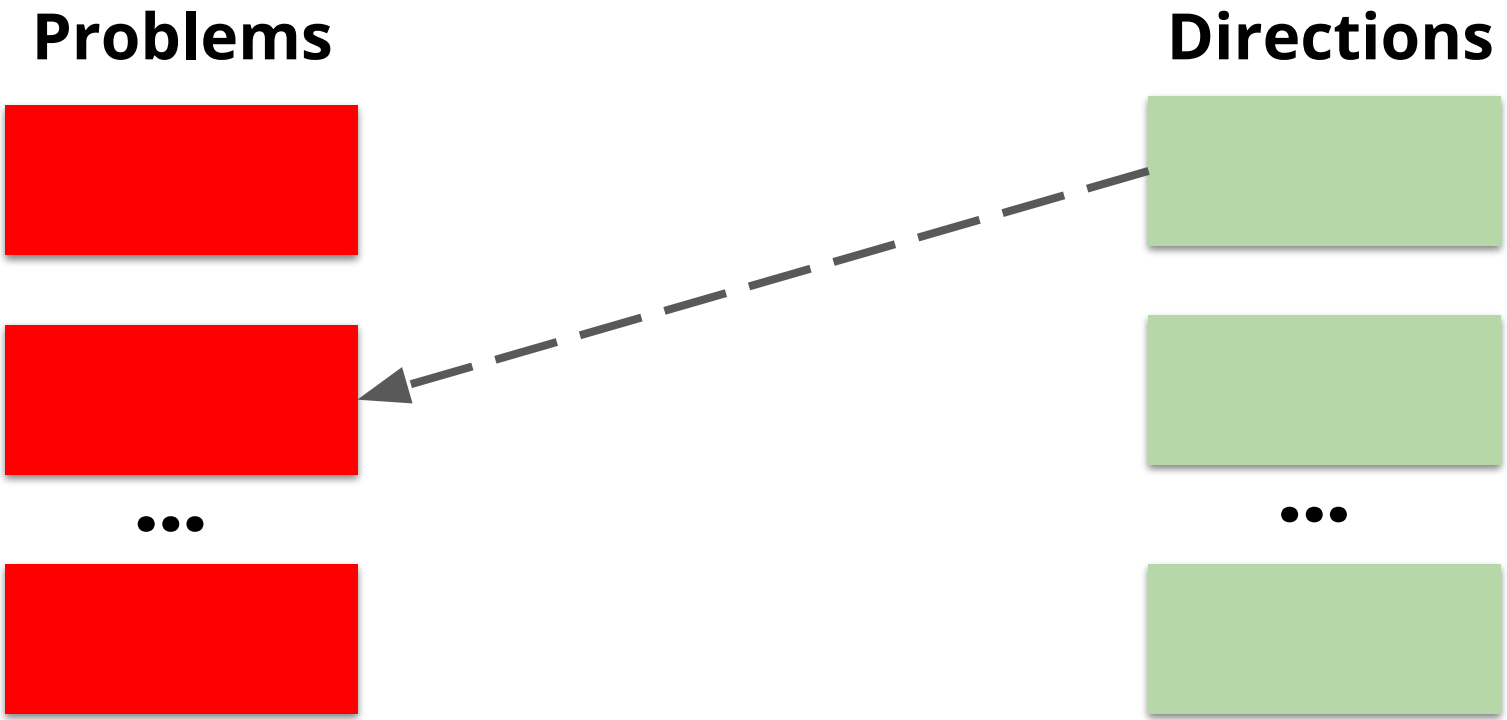
Accelerate the pace of scientific discovery by building agents that help with specific aspects of the scientific discovery process. For example:

- A literature search tool that helps you find relevant papers
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- An experiment builder tool to help you design experiments
- etc.

Autonomous Discovery:

Design agents for end-to-end discovery -- that can, entirely on their own (with minimal human intervention), come up with ideas, hypothesis, design and run experiments, and report the results.

Huge Hypothesis Spaces

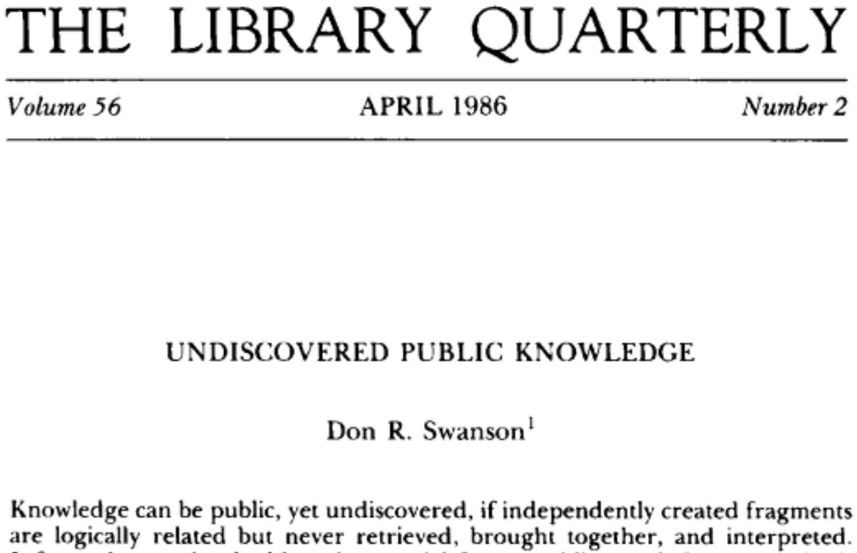


An NLP-based ideator might help generate new ideas based on **existing ideas in the scientific literature** (i.e. by reading papers).

It might work **automatically**, or it might work by asking a human to provide idea generation parameters, and iterate on these (i.e. “**human-in-the-loop**”).

Literature-based discovery (LBD), Don Swanson ‘86

- Leveraging the literature for automatically generating new hypotheses



THE LIBRARY QUARTERLY

Volume 56 APRIL 1986 *Number 2*

UNDISCOVERED PUBLIC KNOWLEDGE

Don R. Swanson[1]

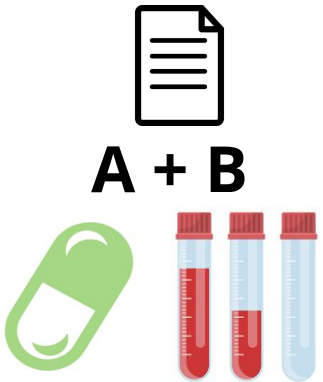
Knowledge can be public, yet undiscovered, if independently created fragments are logically related but never retrieved, brought together, and interpreted.

Literature-based discovery (LBD), Don Swanson ‘86

- **ABC paradigm**

Literature-based discovery (LBD), Don Swanson ‘86

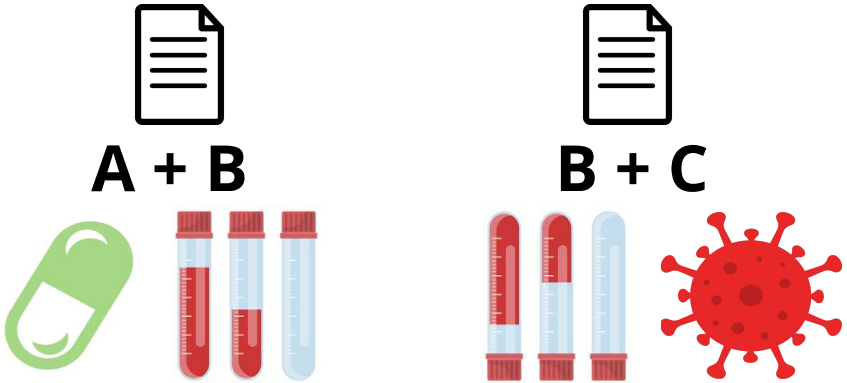
- **ABC paradigm**



*“Drug **A** regulates levels of **B**...”*

Literature-based discovery (LBD), Don Swanson ‘86

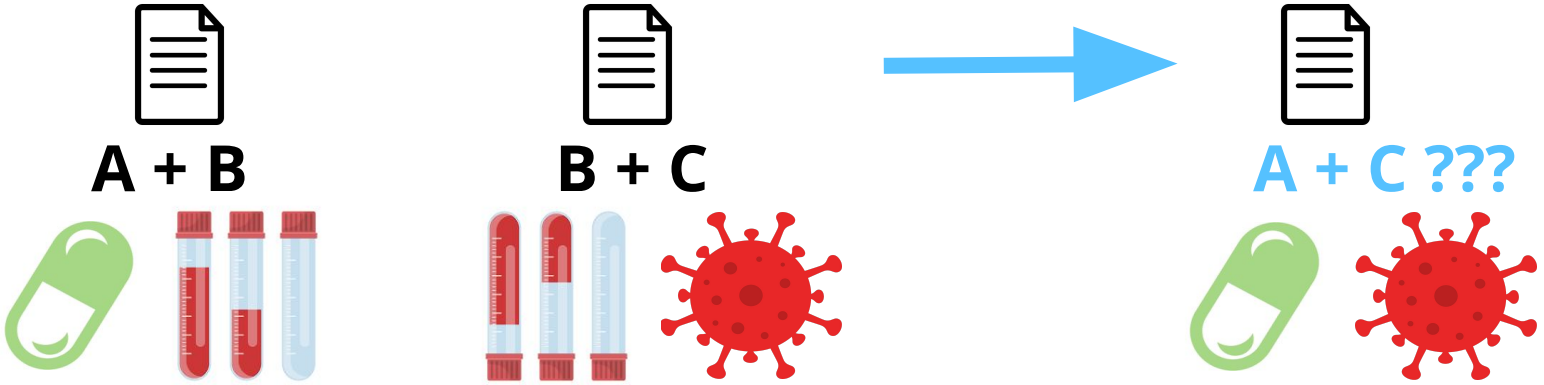
- **ABC paradigm**



*“Disease **C** affects levels of **B**...”*

Literature-based discovery (LBD), Don Swanson ‘86

- **ABC paradigm**



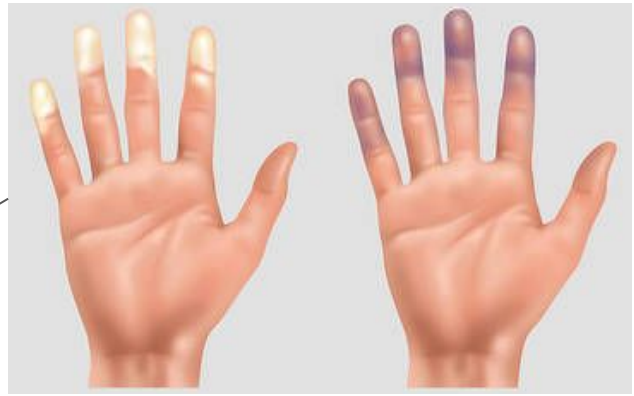
FISH OIL, RAYNAUD'S SYNDROME, AND UNDISCOVERED PUBLIC KNOWLEDGE

DON R. SWANSON*

Fish oil



Raynaud syndrome



Narrowing of arteries supplying blood to skin



Blood viscosity

FISH OIL, RAYNAUD'S SYNDROME, AND UNDISCOVERED PUBLIC KNOWLEDGE

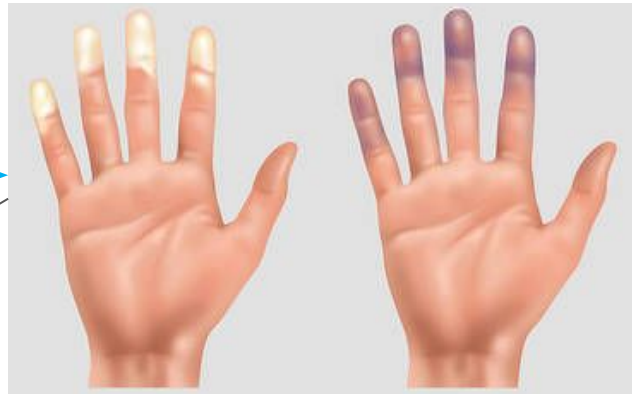
DON R. SWANSON*

Fish oil



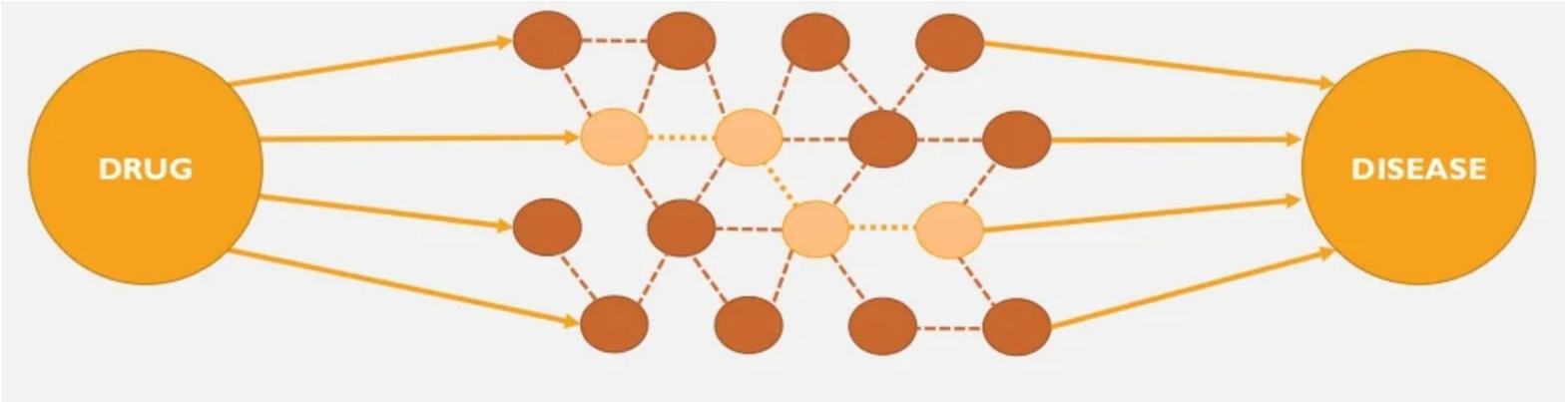
Treatment

Raynaud syndrome



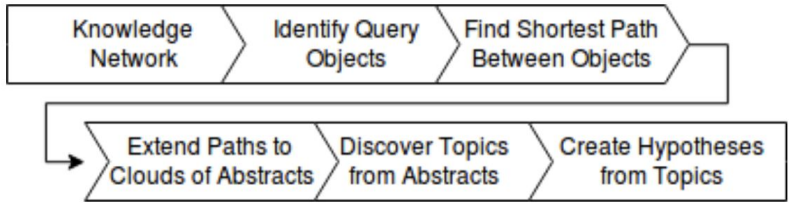
Narrowing of arteries supplying blood to skin

Blood viscosity



MOLIERE: **Automatic Biomedical Hypothesis Generation System**

Justin Sybrandt Michael Shtutman Ilya Safro



Sybrandt et al., KDD 2017

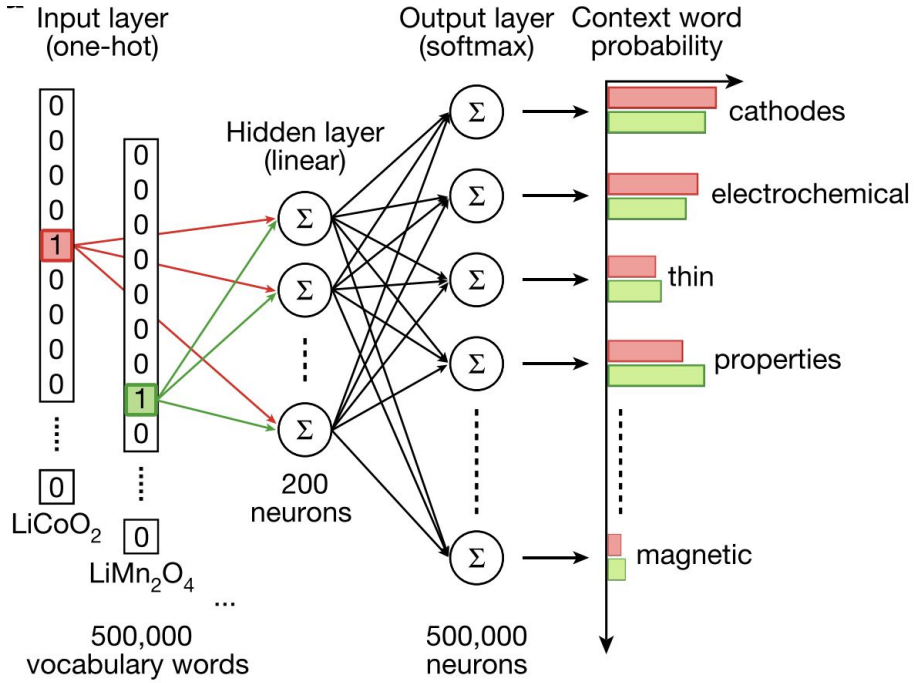
Unsupervised word embeddings capture latent knowledge from materials science literature

Vahe Tshitoyan[1,3]*, John Dagdelen[1,2], Leigh Weston[1], Alexander Dunn[1,2], Ziqin Rong[1], Olga Kononova[2], Kristin A. Persson[1,2], Gerbrand Ceder[1,2]* & Anubhav Jain[1]*

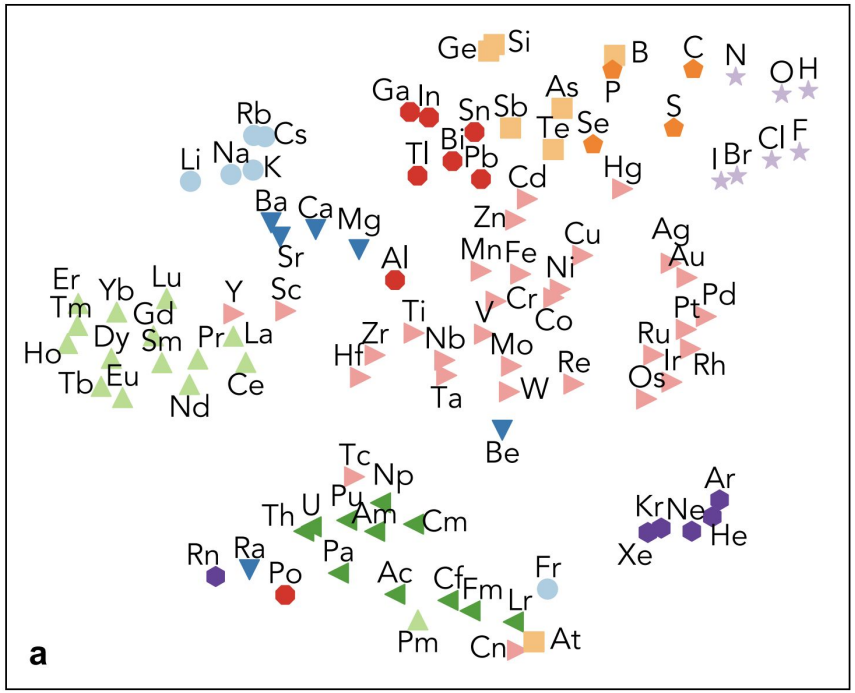
Nature 2019

Unsupervised word embeddings capture latent knowledge from materials science literature

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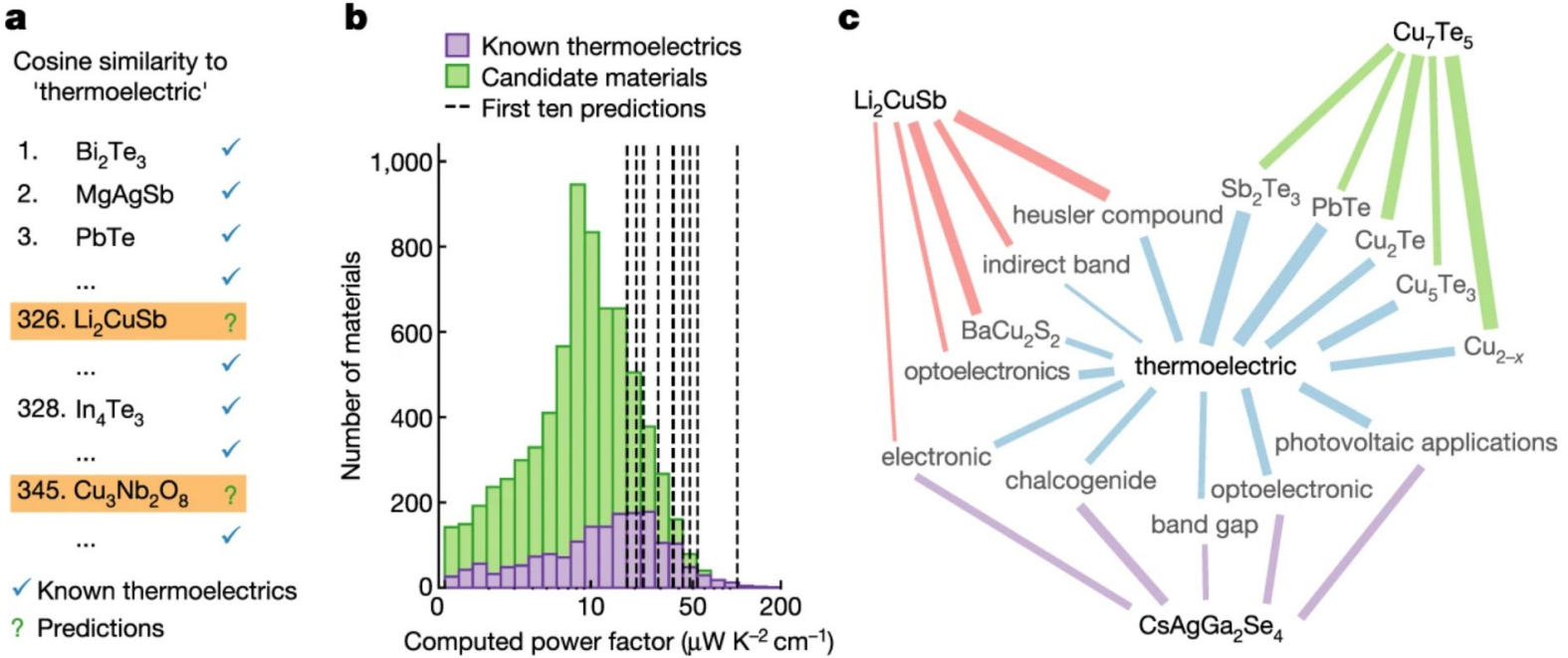
Nature 2019



Extended Data Fig. 1 | Chemistry is captured by word embeddings.

Fig. 2: Prediction of new thermoelectric materials.

From: Unsupervised word embeddings capture latent knowledge from materials science literature



Scimon: Scientific Inspiration Machines Optimized for Novelty

*Q. Wang, D. Downey, H. Ji, **T. Hope** , ACL 2024; arXiv v1: 5/2023*

Scimon: Scientific Inspiration Machines Optimized for Novelty

Background Context

Problems, motivations, focus points...

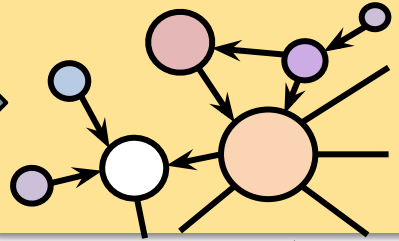
*Q. Wang, D. Downey, H. Ji, **T. Hope** , ACL 2024; arXiv v1: 5/2023*

Scimon: Scientific Inspiration Machines Optimized for Novelty

Background Context

Problems, motivations, focus points…

Inspiration Retrieval

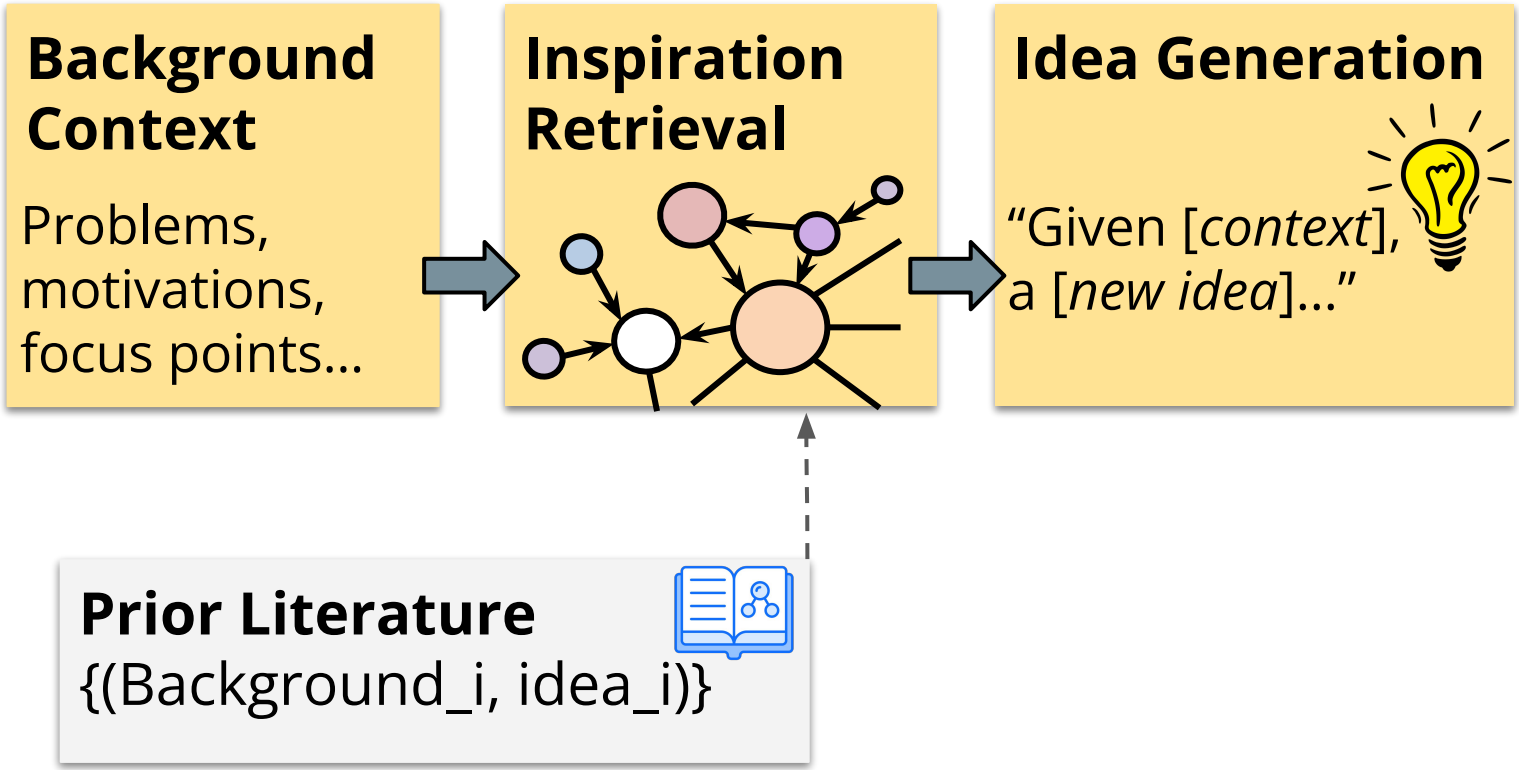


Prior Literature

{(Background_i, idea_i)}

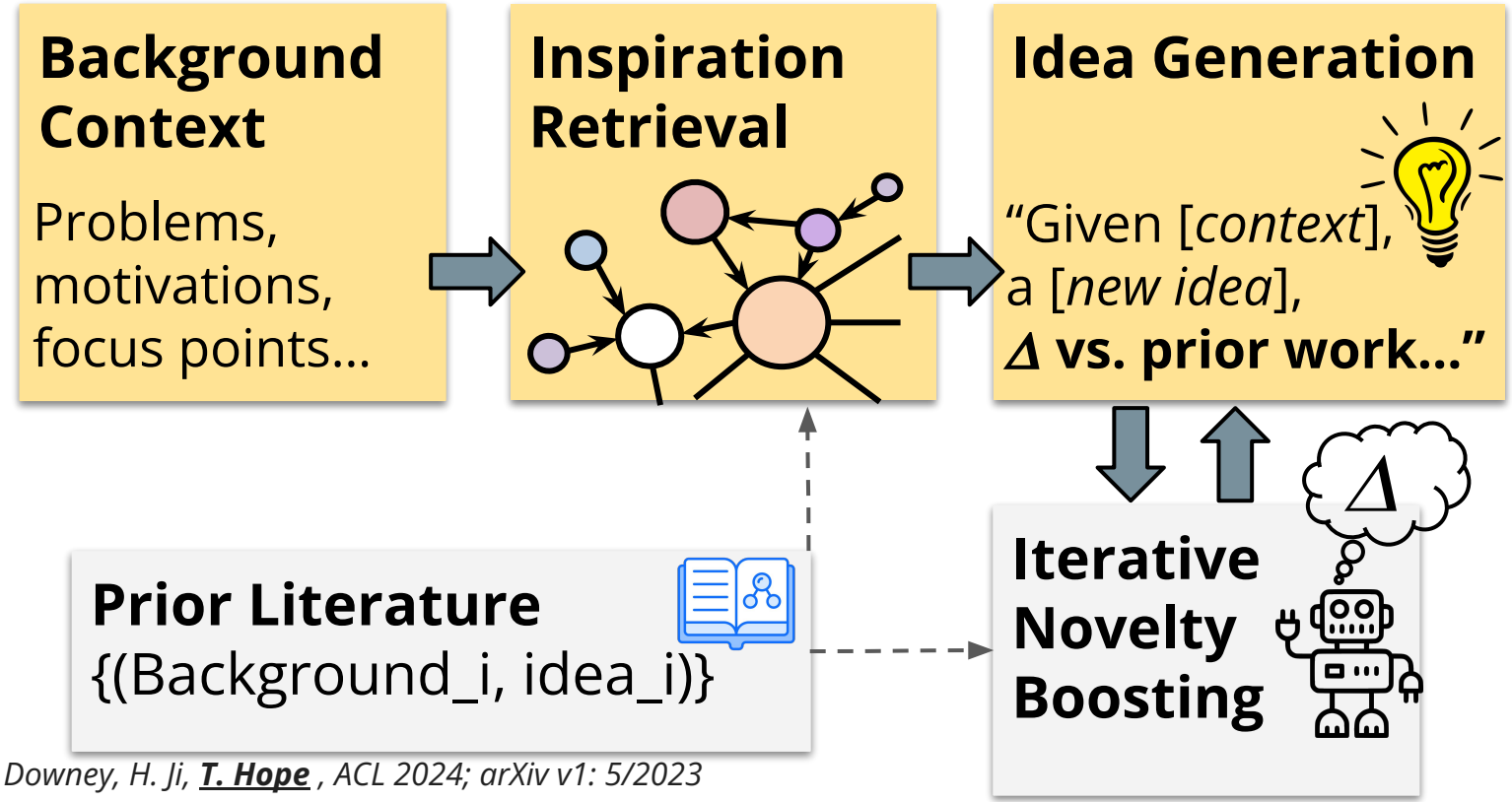
*Q. Wang, D. Downey, H. Ji, **T. Hope** , ACL 2024; arXiv v1: 5/2023*

Scimon: Scientific Inspiration Machines Optimized for Novelty



*Q. Wang, D. Downey, H. Ji, **T. Hope** , ACL 2024; arXiv v1: 5/2023*

Scimon: Scientific Inspiration Machines Optimized for Novelty



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Problem Description → Find Literature Inspirations → Synthesize New Direction

Input

***pre – training with streaming data**, lifelong learning.*
Suggest a new idea

→

Retrieve inspirations

+ continual learning
+ meta fine-tuning
+ memory networks
+ knowledge distillation

→

Meta learning for integrating new information ...
external memory matrix to selectively retrieve information ...
Periodic knowledge distillation...

*Q. Wang, D. Downey, H. Ji, **T. Hope** , ACL 2024; arXiv v1: 5/2023*

Problem/Motivation:
... streaming data of various sources may continuously grow ... requires plms to *integrate the information from all the sources* in a lifelong manner... pre-training on all existing data, such a process is expensive.

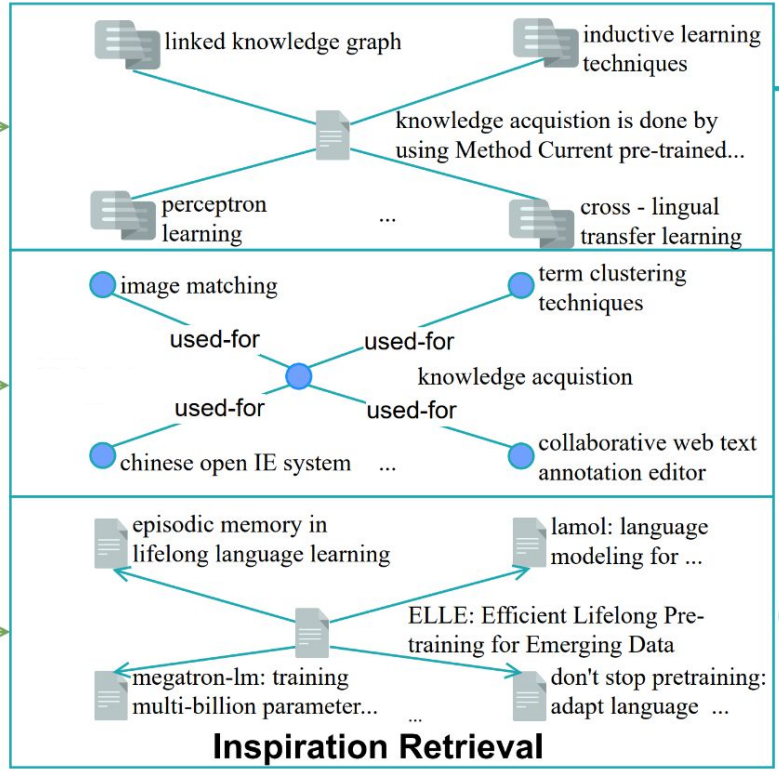
Seed Term:
knowledge acquisition

Input: Background Context

Optional input: *seed concept* to guide hypothesis gen.

Problem/Motivation:
... streaming data of various sources may continuously grow ... requires plms to *integrate the information from all the sources* in a lifelong manner... pre-training on all existing data, such a process is expensive.
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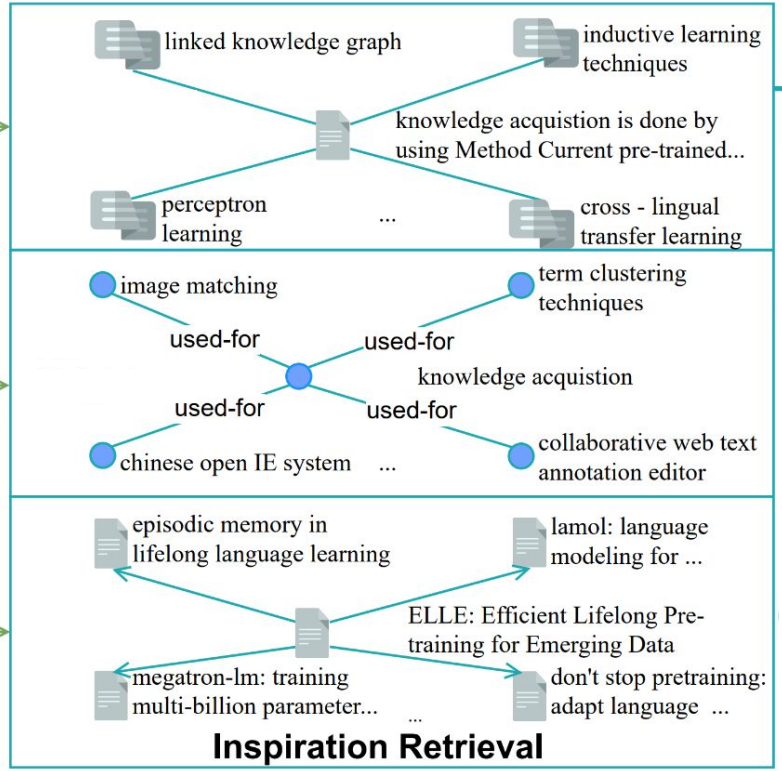
Solutions to past similar problems

Affordances
KG (methods and uses)

Citation-based (assume we know all the references of a future idea paper; including noisy irrelevant refs)

Problem/Motivation:
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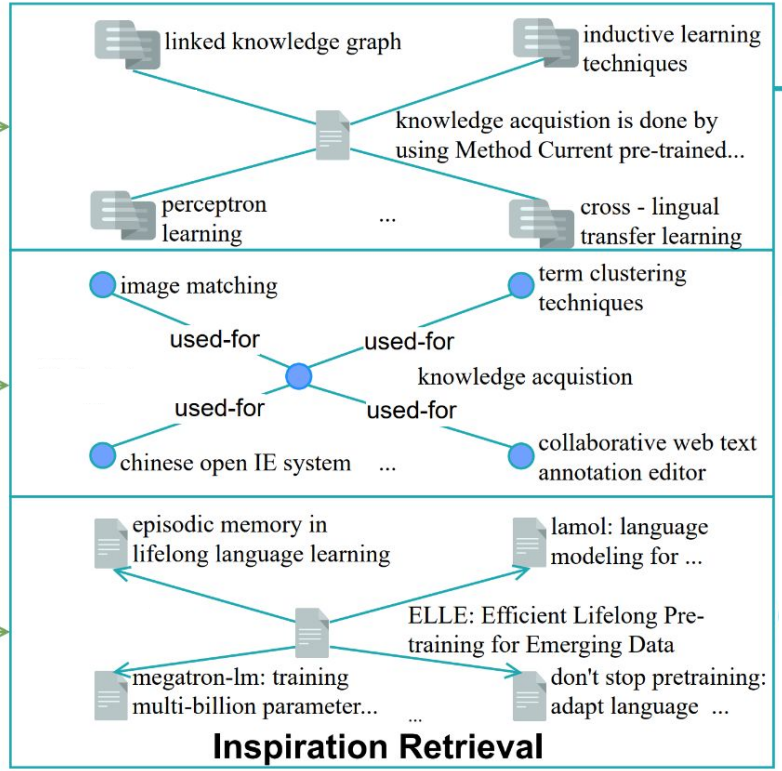


Initial Idea

...continual learning for knowledge acquisition... This approach is more efficient than exhaustive pre-training on all existing data...

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Initial Idea

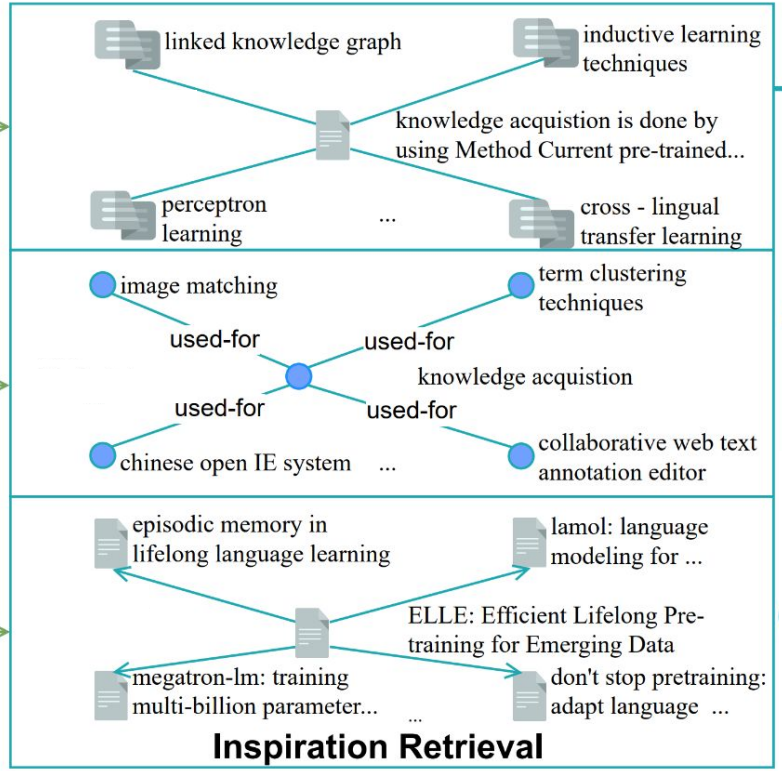
...continual learning for knowledge acquisition... This approach is more efficient than exhaustive pre-training on all existing data...

Retrieved Similar Ideas ↓ Retrieval from Literature Papers

1. Continual learning (CL) aims to enable information systems to learn from a continuous data stream across time...

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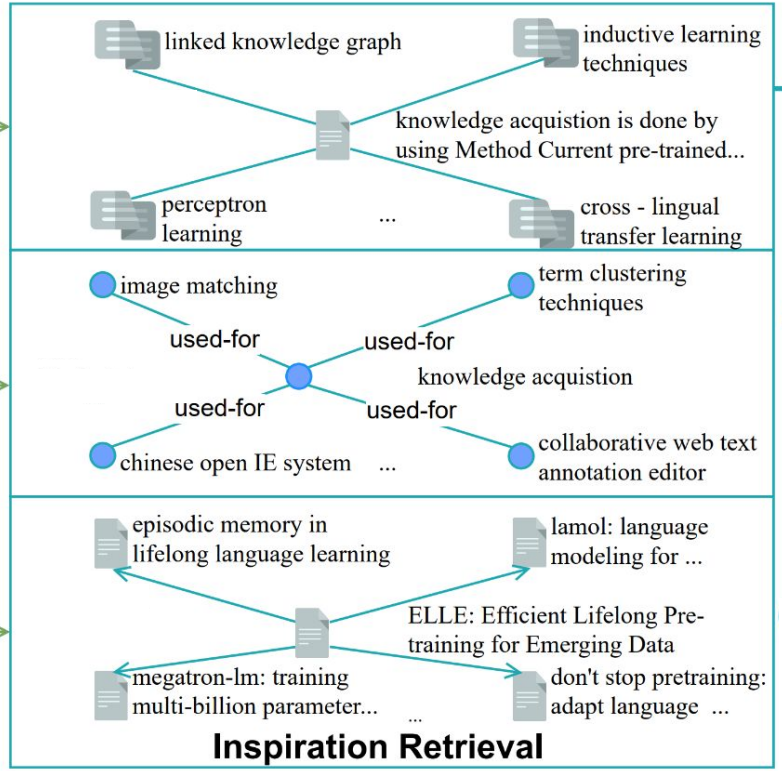
Novelty Threshold Check

1st Novelty Iteration

...a method that combines continual learning with a dynamic knowledge distillation approach for efficient knowledge acquisition ...

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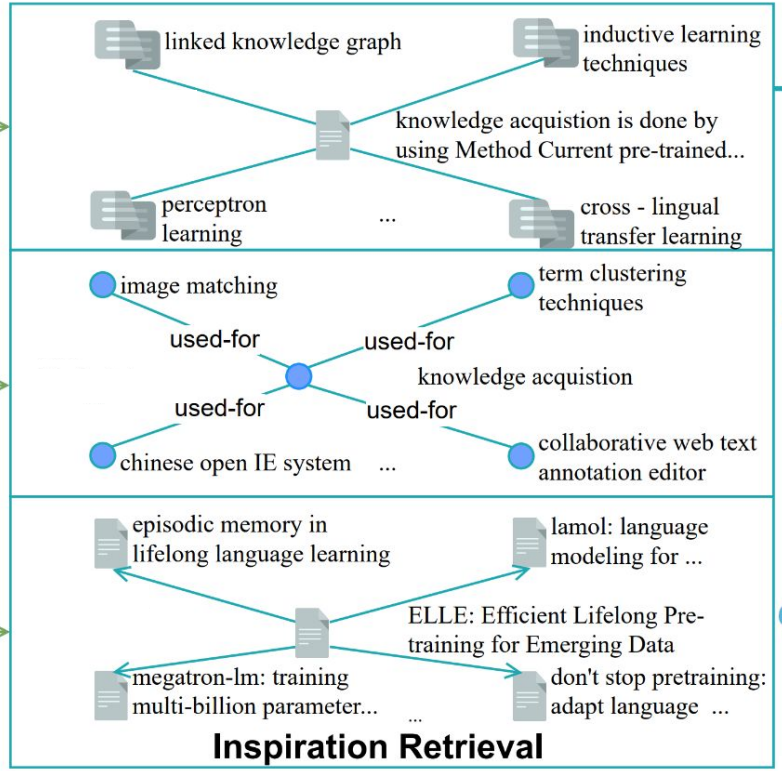
Retrieval from Literature Papers

Retrieved Similar Ideas

1. Different from previous knowledge distillation methods ... student model learns from teacher model for incremental knowledge extraction ...

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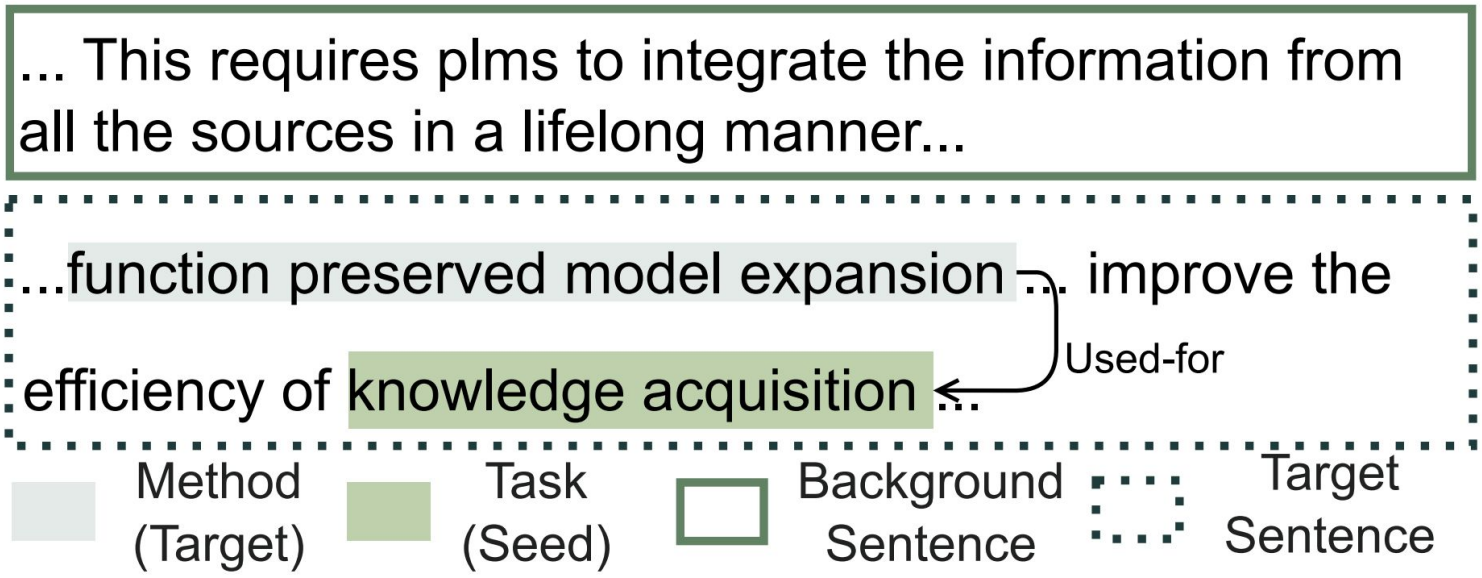
1. Different from previous knowledge distillation methods ... student model learns from teacher model for incremental knowledge extraction ...

Novelty Threshold Check

Final Idea

... a method that leverages memory-augmented neural networks for knowledge acquisition in a lifelong learning scenario...

Extracting Training Data + Inspirations from Past Papers



Encouraging Novelty vs. Background: In Training

... This requires plms to integrate the information from all the sources in a lifelong manner...

In-context negatives

Encouraging Novelty vs. Background: In Training

In-context contrastive objective

Target

$$y^+ = \sigma(\text{Avg}(\mathbf{W}_y \mathbf{h}^+ + \mathbf{b}_y))$$

$$y_k^- = \sigma(\text{Avg}(\mathbf{W}_y \mathbf{h}_k^- + \mathbf{b}_y))$$

In-context negative k

$$\mathcal{L}_{\text{cl}} = \frac{\exp\left(y^+/\tau\right)}{\sum_k \exp\left(y_k^-/\tau\right) + \exp\left(y^+/\tau\right)}$$

Encouraging Novelty vs. Literature: In Inference

Novelty Penalty

$$\gamma_{\text{nov}}(\mathcal{I}, \mathcal{R})$$

Generated idea

Global reference literature

Encouraging Novelty vs. Literature: In Inference

Novelty Penalty

$$\gamma_{\text{nov}}(\mathcal{I}, \mathcal{R})$$

Generated idea

Global reference literature

Minimize $\gamma_{\text{nov}}(\mathcal{I}, \mathcal{R})$

s.t.

$\mathcal{I}$ remains relevant to the background context

Encouraging Novelty vs. Literature: In Inference

Retrieve-Compare-Update

- Retrieve related work
- If similarity < threshold:
  - Instruct model to update in light of related work + initial problem

LLMs for Scientific Inspirations?

- Human evaluation studies
  - Idea quality: **relevant, novel, reasonable, specific, clear, technical detail**
  - Rated also versus the ground truth idea
  - Evaluating **novelty boosting**

1.5X - 2X boost vs. GPT-4 idea ratings
(CS/biomed experts evaluations)

Novelty Boosting Evaluation

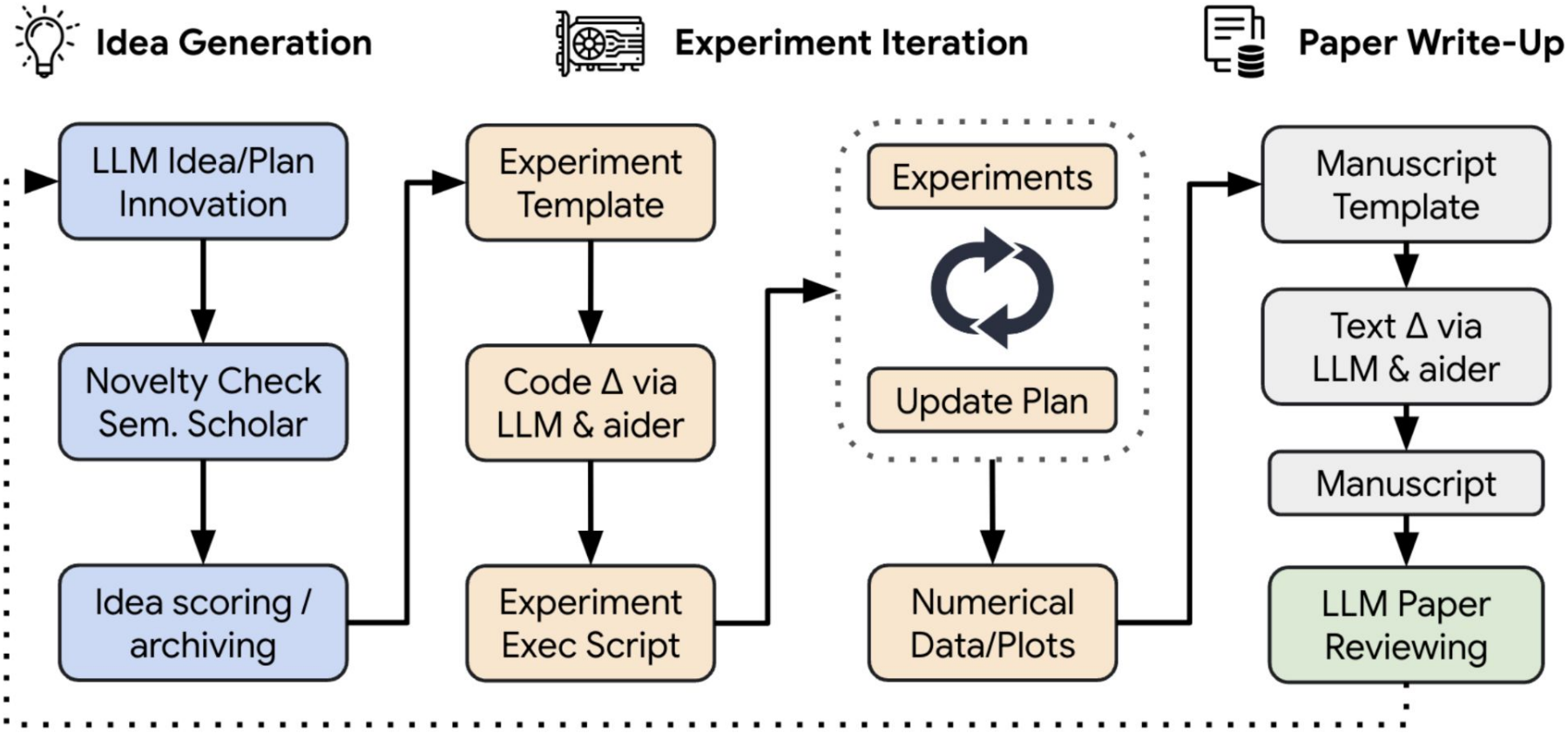
Type of inspiration retrieval (*solutions to past problems, citation-based, knowledge graph*)

	GPT4FS	+SN	+CT	+KG
1st Novelty Δ (%)	+54.4	+55.6	+47.8	+46.7
2nd Novelty Δ(%)	-	+57.8	-	-

1st / 2nd iteration ideas that **gained novelty** over the initial idea

Novelty Boosting Evaluation

	GPT4FS	+SN	+CT	+KG
1st Novelty Δ (%)	+54.4	+55.6	+47.8	+46.7
2nd Novelty Δ(%)	-	+57.8	-	-
1st new terms Δ	+23.1	+22.8	+22.1	+21.9
2nd new terms Δ	-	+21.5	-	-


Idea Generation
Experiment Iteration
Paper Write-Up
LLM Idea/Plan Innovation
Novelty Check Sem. Scholar
Idea scoring / archiving
Experiment Template
Code Δ via LLM & aider
Experiment Exec Script
Experiments
Update Plan
Numerical Data/Plots
Manuscript Template
Text Δ via LLM & aider
Manuscript
LLM Paper Reviewing

LLMs for Scientific Inspirations?

- **<u>Still</u> often generic suggestions, woven with details copied from context**
  - *“NLP with ML algorithms and sentiment analysis for X”*
  - *“Data augmentation and transfer learning for X”*

LLMs for Scientific Inspirations?

- Often generic suggestions, woven with details copied from context
  - *“NLP with ML algorithms and sentiment analysis for X”*
  - *“Data augmentation and transfer learning for X”*
- ***Simple logical modifications to the context***
  - *“high latency” / “efficiency limitations” -> “low latency” / “highly efficient”*

LLMs for Scientific Inspirations?

- Often generic suggestions, woven with details copied from context
  - *“NLP with ML algorithms and sentiment analysis for X”*
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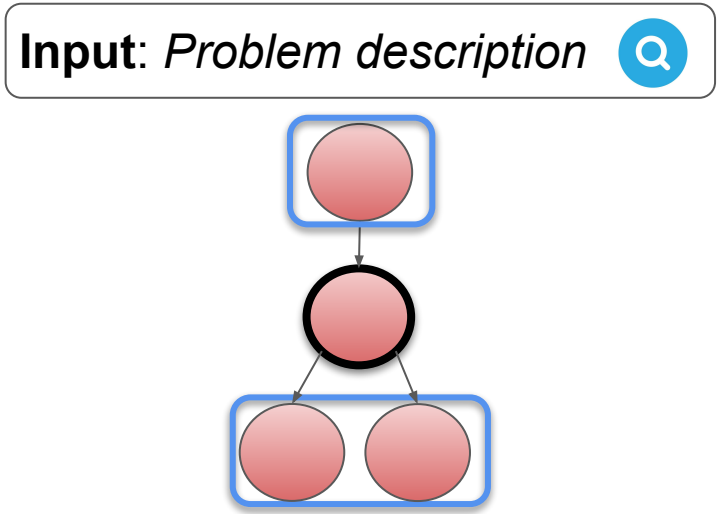
- ***Novelty boosting adds combinations of “buzz concepts” :***
  - *Dynamic/adaptive modeling, graph models and representations, fusion of multiple modalities and sources…*

```
“Dynamic Syntax-Aware Graph Fusion Networks (DSAGFN)”
```

LLMs for Scientific Inspirations?

- Our modelling augmentations produced *promising signal*
  - **Higher technical detail, less generic**
  - **Higher perceived novelty, utility**

... but:
Directions still judged as
far from sufficient in novelty, utility.

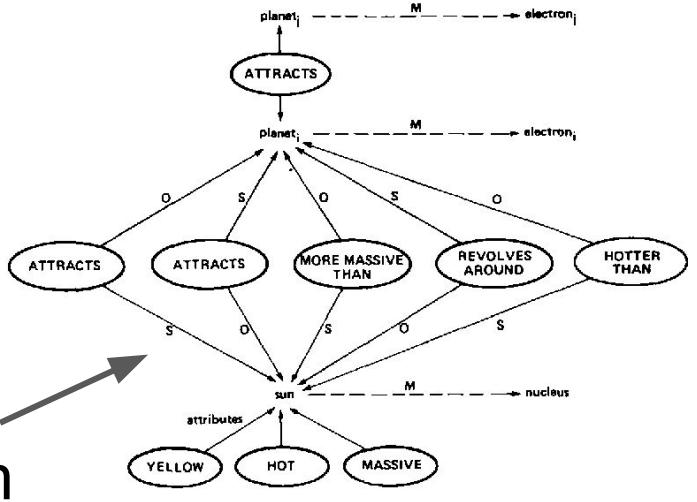


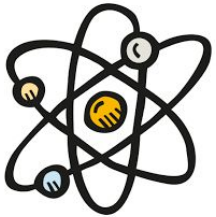
Find novel idea **abstractions, perspectives & interdisciplinary inspirations**

Analogies
for finding cross-domain inspirations

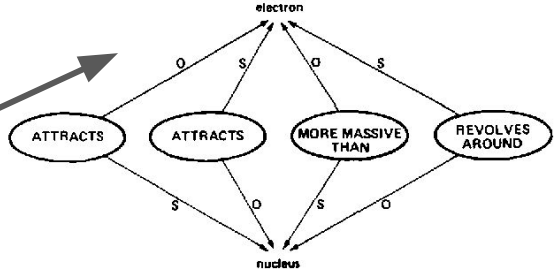


Solar System



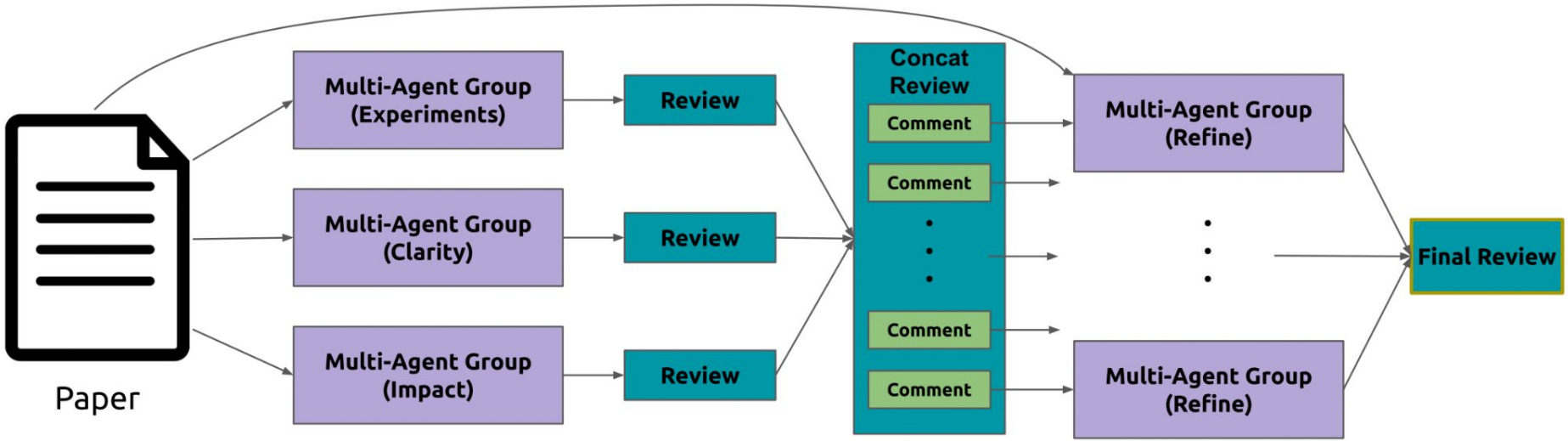


Atom



Critique and Enhance Ideas

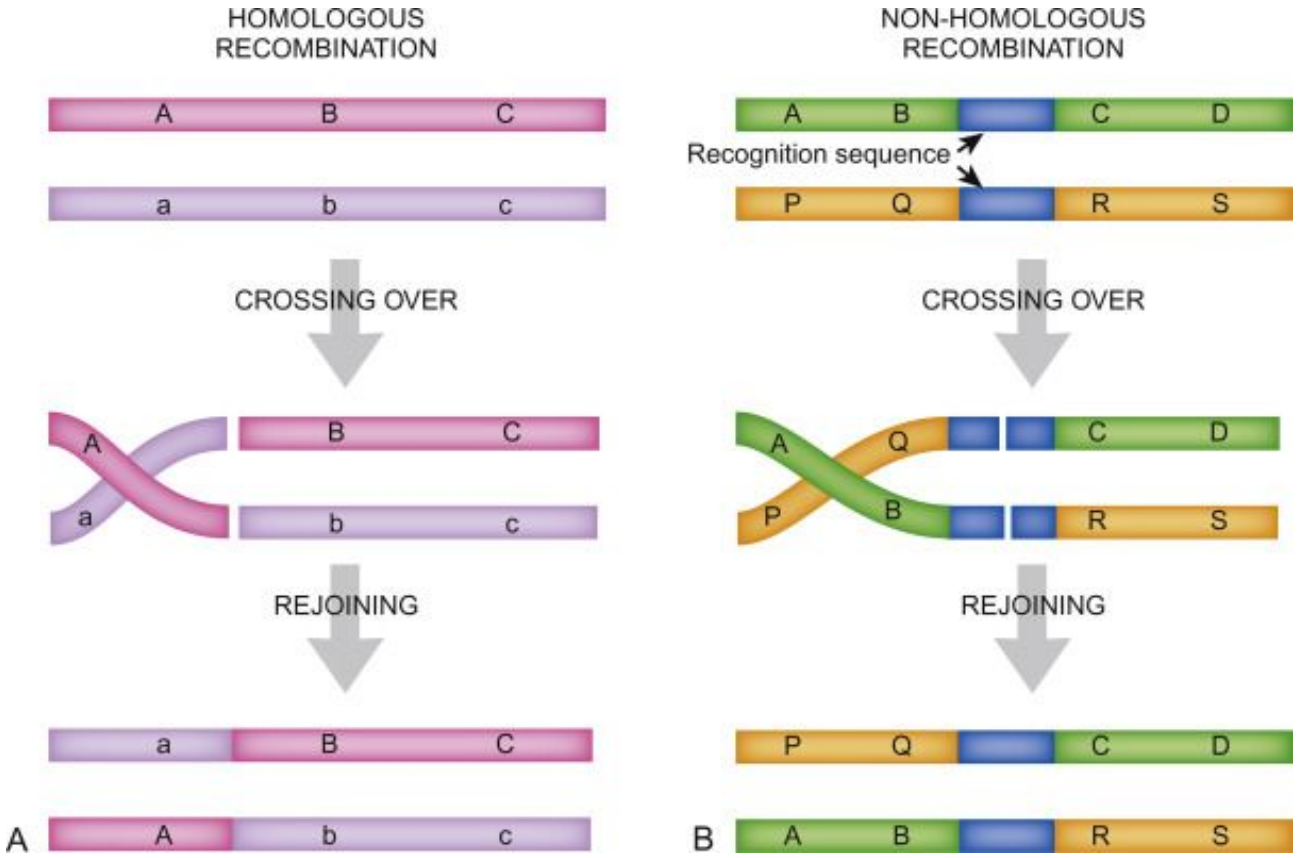
Multiple LLM agents, each focusing on reviewing different aspects

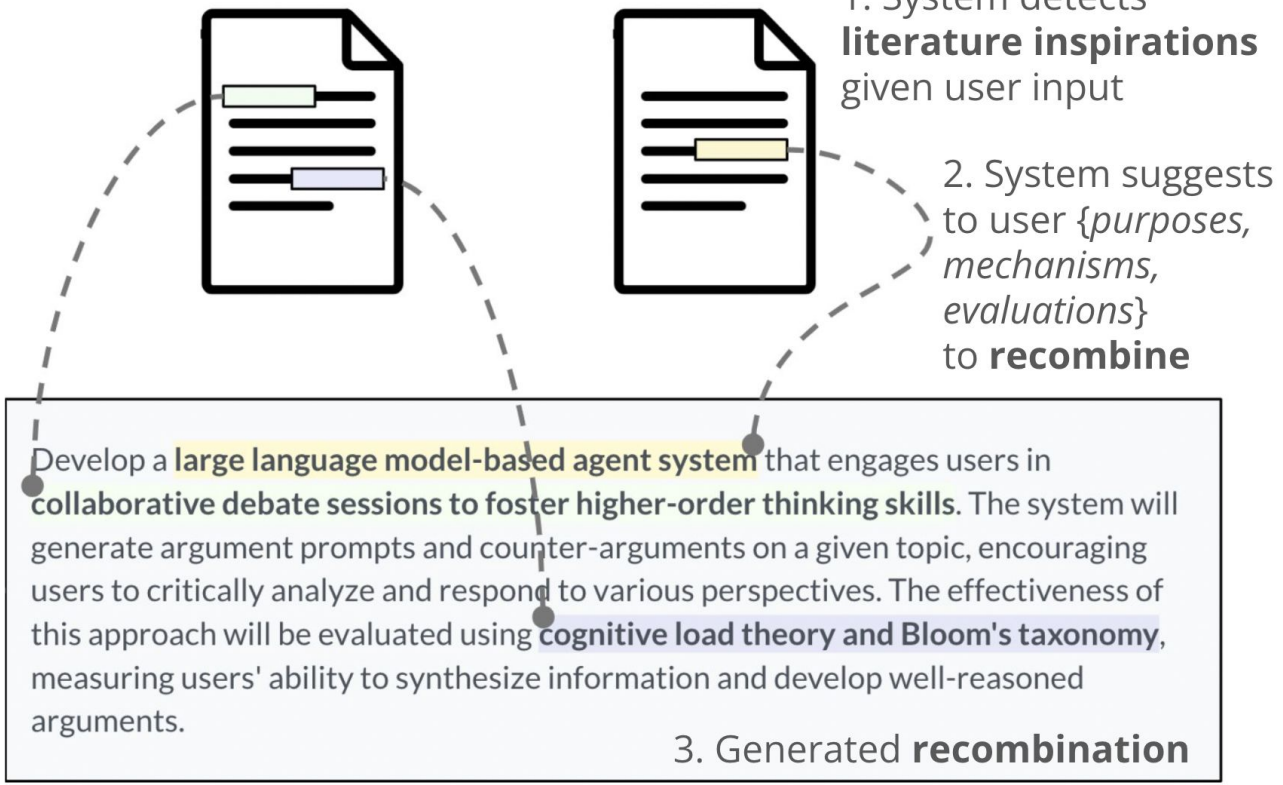


MARG: Multi-Agent Review Generation for Scientific Papers, 2024

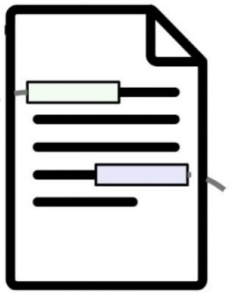
Recombination

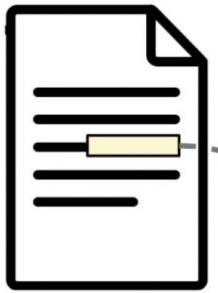



HOMOLOGOUS RECOMBINATION
A B C
a b c
CROSSING OVER
A
a
B C
b c
REJOINING
a B C
A A b c
A
NON-HOMOLOGOUS RECOMBINATION
A B C D
Recognition sequence
P Q R S
CROSSING OVER
A Q C D
P B R S
REJOINING
P Q C D
B A B R S



Scideator: Human-LLM Scientific Idea Generation Grounded in Research-Paper Facet Recombination





1. System detects **literature inspirations** given user input

2. System suggests to user {*purposes, mechanisms, evaluations*} to **recombine**

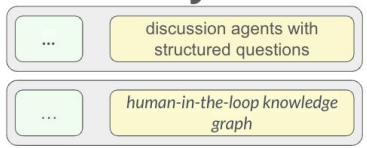
Develop a **large language model-based agent system** that engages users in **collaborative debate sessions to foster higher-order thinking skills.** The system will generate argument prompts and counter-arguments on a given topic, encouraging users to critically analyze and respond to various perspectives. The effectiveness of this approach will be evaluated using **cognitive load theory and Bloom's taxonomy**, measuring users' ability to synthesize information and develop well-reasoned arguments.

3. Generated **recombination**

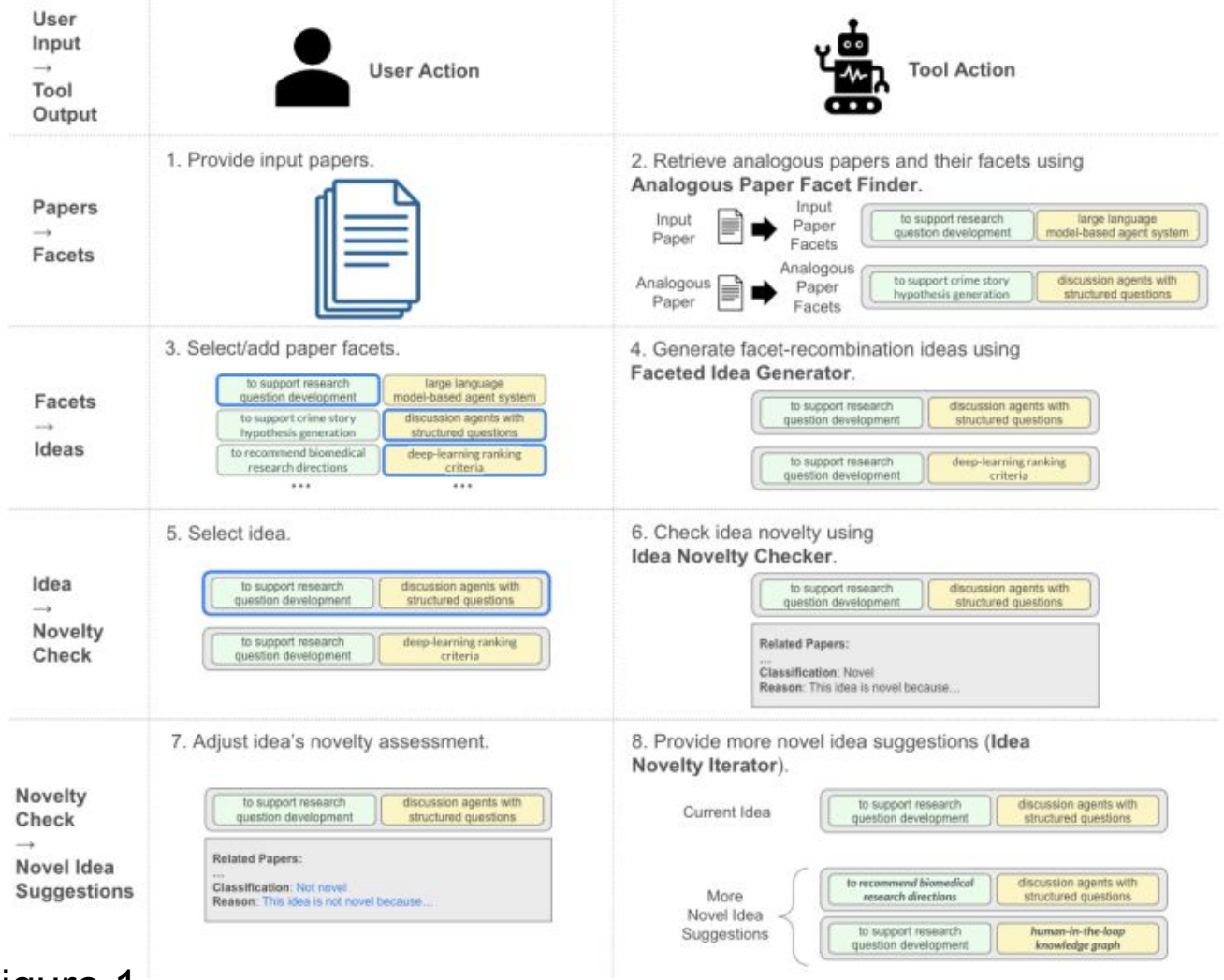
```
Related Papers:
[...]
Classification: Not novel
Reason: This idea is not novel
because...
```

4. **Novelty** Assessor

5. **Novelty** Iterator



More Novel Idea Suggestions

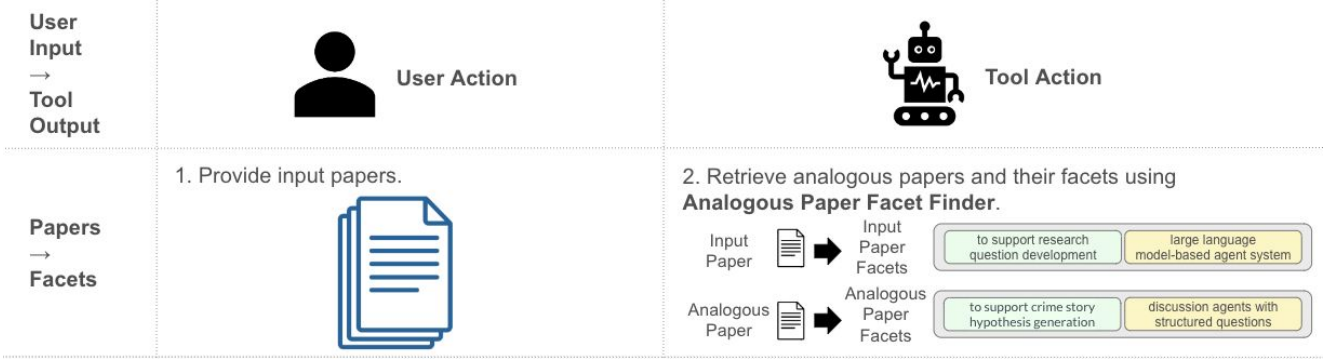


Extract components of papers

Combine components in novel fashion

Check whether the ideas are novel

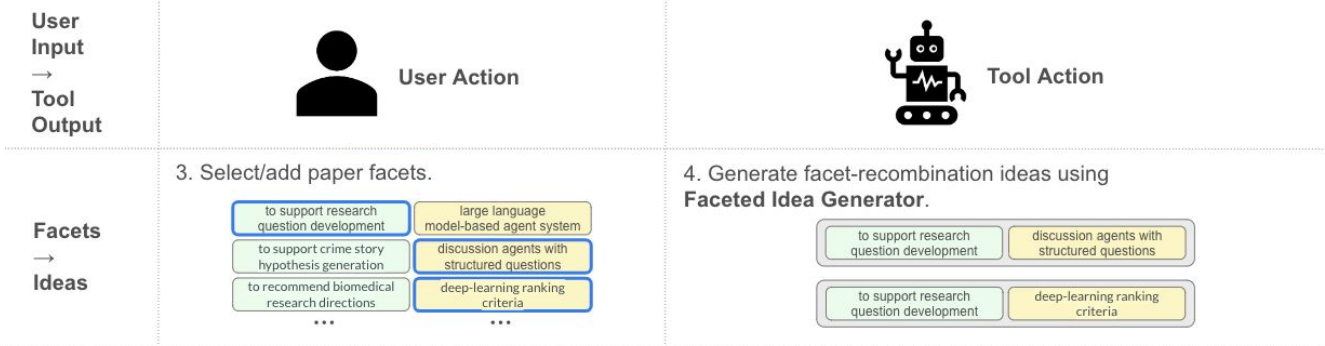
Figure 1



User provides a set of input papers that they would like to use as a starting point for ideation

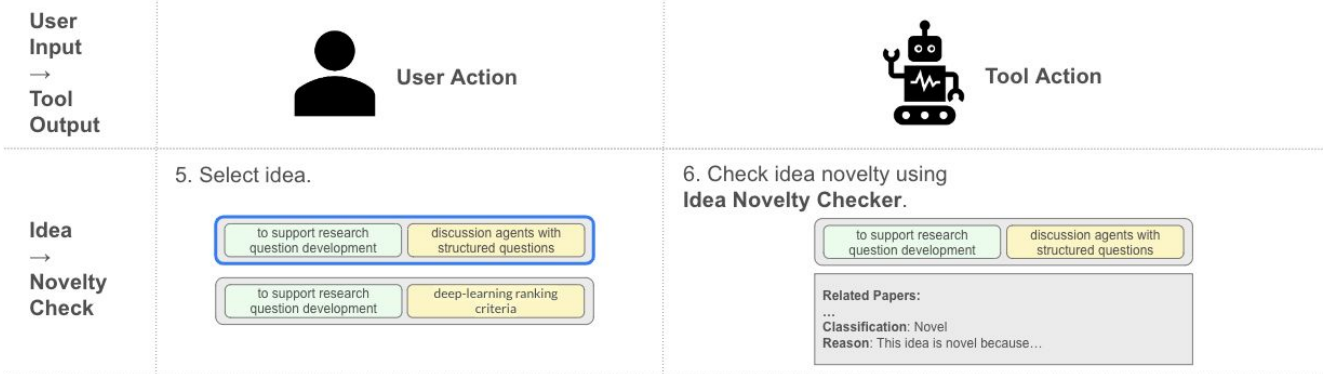
The tool responds by retrieving **analogous** papers to the input papers and extracting facets (*purpose, mechanism*) from the input and analogous papers.

Analogous: papers with similar purpose but different mechanism, or vice versa



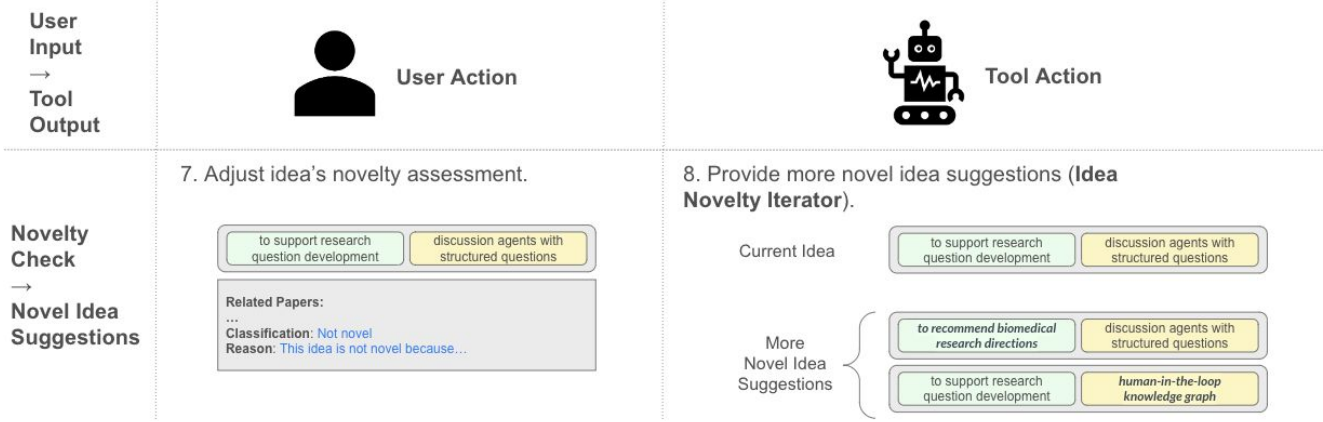
User then selects paper facets as well as adds their own facets for which they want to generate ideas.

The tool **recombines** these selected facets into ideas with one purpose and one mechanism. If the user did not select any facets of a certain type, the tool selects one for them



The user selects an idea to assess for novelty

The tool classifies the idea as "novel" or "not novel" and provides a short reason as to why.



The user reviews the novelty classification and adjusts it if they disagree

If the idea is determined to be "not novel," the tool provides suggestions for more novel ideas that replace one of the initial idea's facets.

Select facets to generate new ideas. Click a facet to filter the generated ideas below.

Purpose Used	Mechanism Used	Evaluation Used
☐ to support research question development (input) ?	☐ large language model-based agent system (input) ?	☐ controlled experiment with researchers (input) ?
☐ to enhance scientific experimentation (near) ?	☐ large language models in experimentation (near) ?	☐ observations and evaluations (near) ?
☐ to accelerate research ideation (near) ?	☐ agent-based interactive system (near) ?	☐ end-user evaluations (near) ?
☐ to enhance ai story generation	☐ multi-agent communication pipe-	☐ api and manual assessments (near) ?
Enter your own purpose	Enter your own mechanism	Enter your own evaluation
Add	Add	Add

Leave blank to generate any new facets, or enter query (up to 5 words) for specific new facets.

Generate More Facets

Generate New Ideas

Explore your ideas. Click an idea's magnifying glass to evaluate it for novelty. Save ideas that you think are both novel and interesting to think about further.

Using a **large language model-based agent system to enhance domain-specific data annotation** in the context of developing research questions. The system will leverage active learning to iteratively present the most informative research questions to users, allowing them to refine and annotate these questions with expert insights. This iterative feedback loop will not only improve the quality of research questions but also foster deeper engagement and creativity. The effectiveness of this approach will be assessed through a **controlled experiment with researchers**, examining both the quality and quantity of generated research questions and user satisfaction. Collapse

Employing **transfer and active learning strategies to support research question development**. This approach will use pre... Expand

Utilize a large language model-based agent system to identify genetic biomarkers in cancer subtypes. The L... Expand

Employ knowledge graph-based recommendation algorithms to support research question development. By levera... Expand

In future work, a **large language model-based agent system** could be developed **to enhance scientific experimentation**. Th... Expand

Future research could explore using **ai-human collaboration to support research question development**. By creating a col... Expand

Develop a **large language model-based agent system to enhance user experience in research question development** by incor... Expand

Apply **user-centered design and design thinking methodologies** to the development of a large language model-based agent system aimed Expand

Enter your own idea

0/1000 characters

Add

IDEA

Developing a large language model-based agent system to complement human creativity with AI in academic research question formulation could lead to more innovative and interdisciplinary research inquiries. The system could present researchers with diverse perspectives and out-of-the-box thinking prompts, fostering a richer brainstorming environment. The effectiveness of this approach can be evaluated using content analysis of student perspectives, where students' feedback and the diversity of generated research questions would be analyzed to assess creativity and comprehensiveness.

FACETS

to complement human creativity with ai (near) ? | large language model-based agent system (input) ? | content analysis of student perspectives (near) ?

RELATED PAPERS

Paper 0: How AI Processing Delays Foster Creativity: Exploring Research Question Co-Creation with an LLM-based Agent

Y. Liu | S. Chen | H. Cheng | +4 authors | Y. Huang • Computer Science • International Conference on Human Factors in Computing Systems • 2023-10-09

Developing novel research questions (RQs) often requires extensive literature reviews, especially... Expand

Paper 1: Enhancing AI Creativity: A Multi-Agent Approach to Flash Fiction Generation with Small Open-Source Models

A. Wang | B. Berwyn | J. Zhao | R. Li

In this work, we explore the potential of multi-agent communication to enhance the creative... Expand

Paper 9: The Creative Value of Bad Ideas – A Computational Model of Creative Ideation

R. Sosa | J. Gero • Psychology • CAADRIA proceedings

This paper analyses two ideation principles: idea accessibility and idea connectivity. Access... Expand

NOVELTY EVALUATION

Rating of Idea Compared to Related Papers:

novel

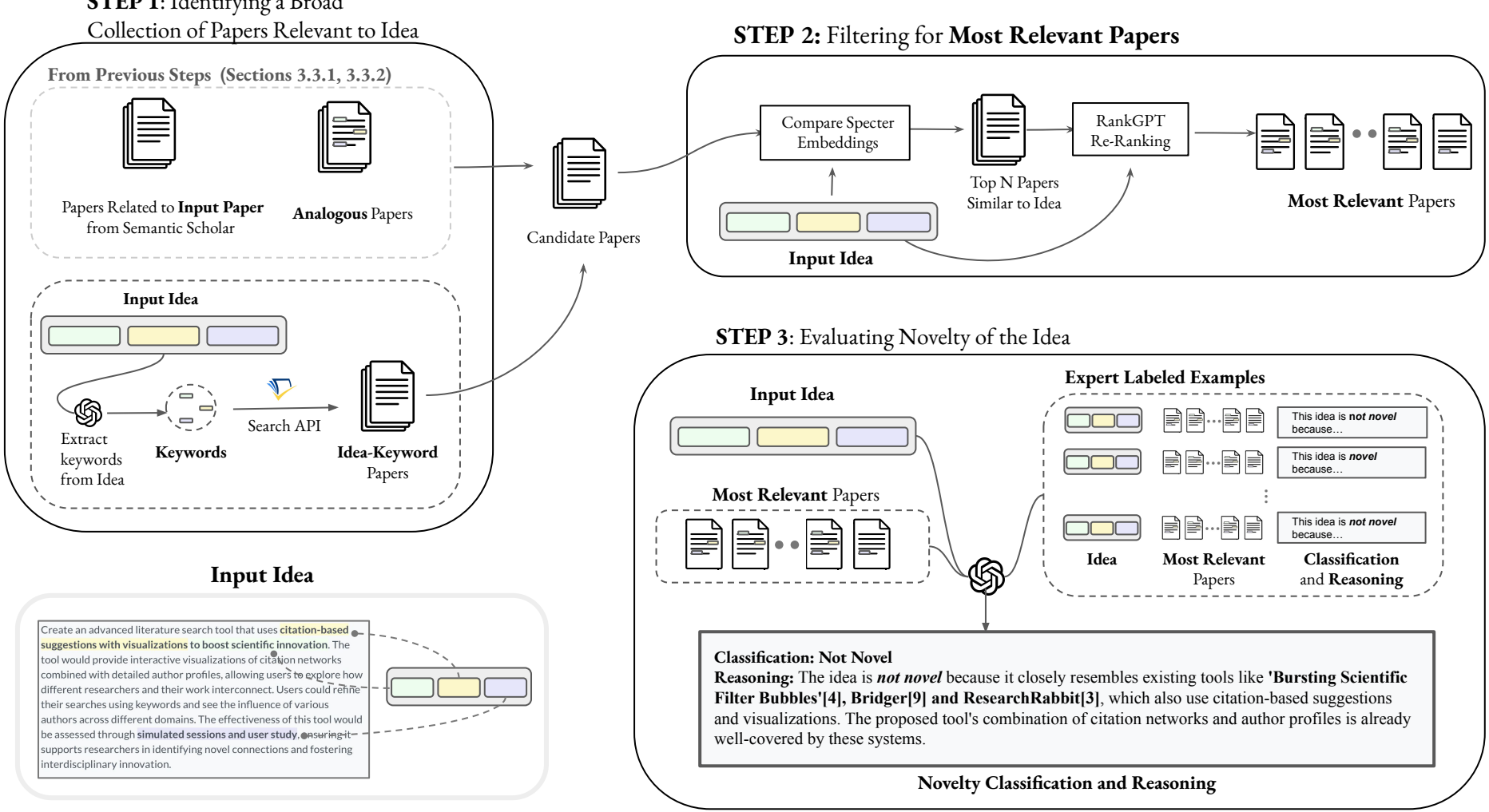
Reason:

The idea is novel because it introduces a large language model-based agent system specifically designed to complement human creativity in academic research question formulation, which is not explicitly addressed in the related papers. For instance, CoQuest [0] focuses on RQ generation but does not emphasize interdisciplinary research inquiries or content analysis of student perspectives. Similarly, other papers like Acceleron [2] and AGATHA [6] explore AI's role in research ideation and hypothesis generation but do not target the specific goal of fostering innovative and interdisciplinary research through diverse perspectives and out-of-the-box thinking prompts.

NEW IDEA SUGGESTIONS

Get suggestions for updating your idea to be novel compared to all the related papers.

No suggestions provided. The idea is judged to be novel compared to relevant papers. If you disagree, change the rating to "not novel", adjust the reason as you please, and refresh the new idea suggestions.


STEP 1: Identifying a Broad Collection of Papers Relevant to Idea
From Previous Steps (Sections 3.3.1, 3.3.2)
Papers Related to Input Paper from Semantic Scholar
Analogous Papers
Input Idea
Extract keywords from Idea
Keywords
Search API
Idea-Keyword Papers
Candidate Papers
STEP 2: Filtering for Most Relevant Papers
Compare Specter Embeddings
Top N Papers Similar to Idea
RankGPT Re-Ranking
Most Relevant Papers
Input Idea
STEP 3: Evaluating Novelty of the Idea
Input Idea
Most Relevant Papers
Expert Labeled Examples
This idea is not novel because…
This idea is novel because…
This idea is not novel because…
Idea
Most Relevant Papers
Classification and Reasoning
Classification: Not Novel
Reasoning: The idea is not novel because it closely resembles existing tools like 'Bursting Scientific Filter Bubbles'[4], Bridger[9] and ResearchRabbit[3], which also use citation-based suggestions and visualizations. The proposed tool's combination of citation networks and author profiles is already well-covered by these systems.
Novelty Classification and Reasoning
Input Idea
Create an advanced literature search tool that uses citation-based suggestions with visualizations to boost scientific innovation. The tool would provide interactive visualizations of citation networks combined with detailed author profiles, allowing users to explore how different researchers and their work interconnect. Users could refine their searches using keywords and see the influence of various authors across different domains. The effectiveness of this tool would be assessed through simulated sessions and user study, ensuring it supports researchers in identifying novel connections and fostering interdisciplinary innovation.

Example 2

Idea: Develop a **systematic review-based framework** designed **to align LLM evaluation with human preferences**, ensuring that evaluation criteria are continuously refined based on comprehensive reviews of user feedback and emerging model behaviors. This framework will utilize **content analysis of user interactions and feedback** to identify patterns and areas of improvement. The effectiveness of this framework will be assessed through a **qualitative study** involving iterative cycles of user feedback and criteria refinement.

Most Relevant Papers:

(1) EvalLM: Interactive Evaluation of Large Language Model Prompts on User-Defined Criteria
(2) HUMANELY: HUMAN EVALUATION OF LLM YIELD, USING A NOVEL WEB BASED EVALUATION TOOL
(3) Evaluation of Code Generation for Simulating Participant Behavior in Experience Sampling Method by Iterative In-Context Learning of a Large Language Model
(4) Human-Centered Evaluation and Auditing of Language Models
(5) Aligning Model Evaluations with Human Preferences: Mitigating Token Count Bias in Language Model Assessments
(6) Who Validates the Validators? Aligning LLM-Assisted Evaluation of LLM Outputs with Human Preferences
(7) Human-Centered Design Recommendations for LLM-as-a-judge
(8) CheckEval: Robust Evaluation Framework using Large Language Model via Checklist
(9) Discovering Language Model Behaviors with Model-Written Evaluations
(10) Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models

Reasoning: The idea is **not novel** because it closely resembles existing frameworks like EvalLM[1] and HumanELY[2], which already align LLM evaluations with human preferences using user-defined criteria and human feedback.

Research Idea

10 Most Relevant Papers (Title, and Abstract)

Novelty Reasoning

Models	Accuracy	Precision	Recall	F1	Cohen Kappa
AI Scientist [30]	0.47	0.55	0.53	0.44	0.05
AI Researcher [41]					
- GPT-4o	0.78	0.81	0.74	0.75	0.52
- Claude-3-5-sonnet	0.56	0.63	0.61	0.56	0.19
Scideator (Ours)	0.81	0.84	0.78	0.79	0.59